Commercial Real Estate and Credit Misallocation

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Abstract

This paper accounts for the heightened sensitivity of Commercial Real Estate (CRE) investors to credit conditions and examines its macroeconomic implications. I first document that CRE investors hold a large share of capital and face looser credit standards than other non-financial firms, owing to the higher pledgeability of their assets. I then incorporate CRE investors into a two-sector general equilibrium model with heterogeneous borrowing constraints. This feature generates sectoral misallocation: capital accumulated by CRE investors yields lower expected returns than capital held by other non-financial firms. I show that this misallocation is substantial, both in steady state and over the credit cycle, by calibrating the model to French macro and micro data. Because sectoral capital stocks are imperfect substitutes, policies that reallocate credit from CRE investors toward other non-financial firms raise welfare, particularly when implemented during the boom phase of a credit cycle.

Keywords: *real estate; misallocation; credit cycle; macroprudential policy.*

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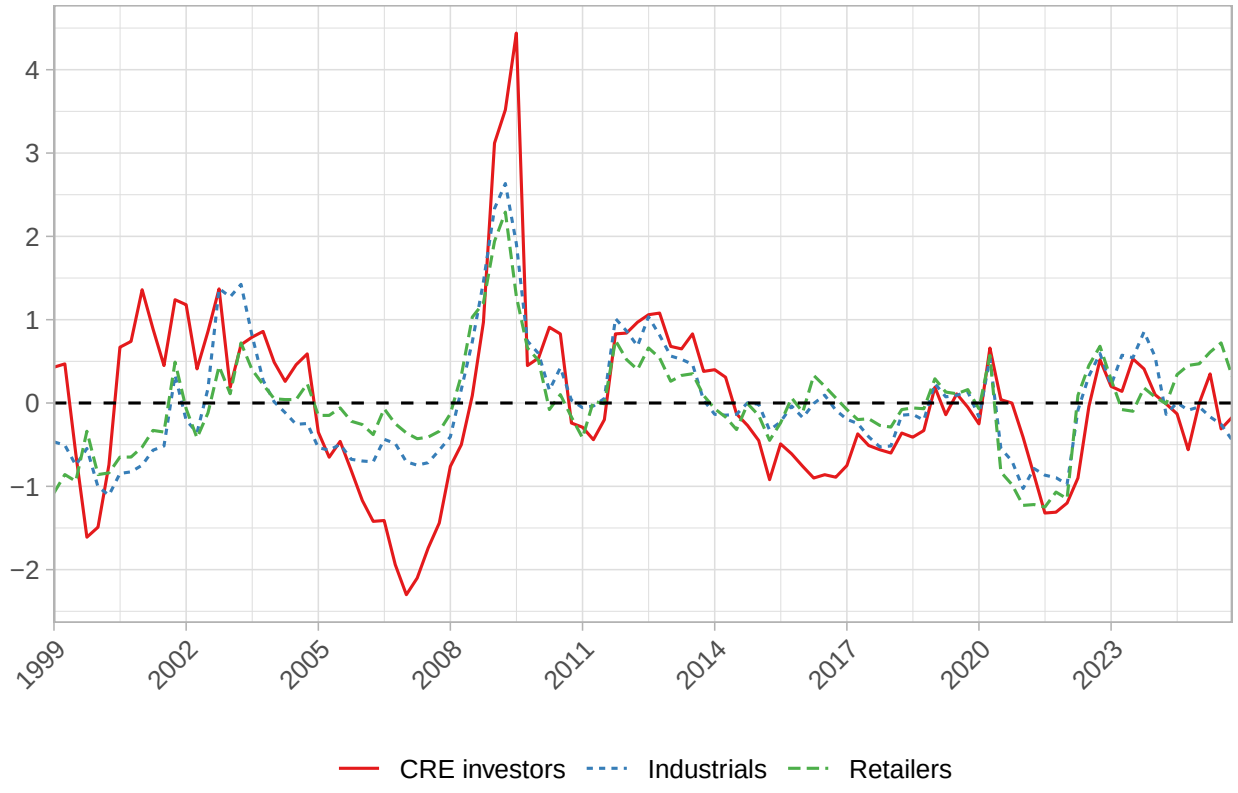
1 Introduction

The 2008 Global Financial Crisis (GFC) reminded that credit supply shocks can be important drivers of the business cycle (Jermann and Quadrini, 2012). A few years later, faced with the lower bound on their conventional policy rates, central banks implemented credit supply policies (Darracq-Paries and De Santis, 2015). Assessing the effects of credit supply shocks—whether they originate in the private sector or result from public policies—is thus a central issue. In particular, heterogeneous exposures to such shocks have gathered increased interest. For instance, Commercial Real Estate (CRE) investors, *i.e.*, firms supplying real estate (shopping centers, offices, etc.) to other firms through a rental market, stand out as particularly sensitive to credit conditions. Figure 1 plots the dividend yield of non-financial sectors in the Eurozone over the past decades and shows that CRE investors dramatically outperformed other non-financial issuers in the run-up to the GFC: loose credit conditions pushed CRE valuations up and yields down (Duca and Ling, 2020). Yields differential significantly widened again in 2014-2017, when the European Central Bank began implementing unconventional monetary policies (Berg et al., 2023).

Figure 1: Sectoral dividend yields in Europe (deviation from historical average)



Source: LSEG Datastream. Sectors are based on the Industry Classification Benchmark (ICB).

This paper accounts for the heightened sensitivity of CRE investors to credit conditions and examines its macroeconomic implications. Leveraging French firm-level data, I document that CRE investors hold a large share of capital and face looser credit standards than other non-financial firms, owing to the higher pledgeability of their assets. I then introduce stylized CRE investors in a general equilibrium model with two production sectors characterized by heterogeneous borrowing constraints. These investors supply CRE to other non-financial firms through a rental market, and benefit from looser credit conditions to accumulate capital. I show that this feature creates heterogeneous exposure to credit supply shocks, and sectoral misallocation: capital accumulated by CRE investors displays lower expected returns than capital accumulated by other non-financial firms. I calibrate the model to French macro and micro data, then show that misallocation is large, both at the steady state and over the credit cycle. Because capital owned by each sector are imperfect substitutes, policies that reallocate credit from CRE investors towards other non-financial sectors raise welfare, particularly when implemented during the boom phase of a credit cycle. This is a topical concern concern for macroprudential authorities, who have been closely scrutinizing CRE investors over the past few years.[1]

I proceed in four steps. I first present some stylized facts on CRE investors. To do so, I construct a dataset documenting French firms' balance sheet structure and credit conditions. The first source for this dataset is Fiben, a database gathering financial accounts, financial links, and credit ratings of a large sample of French non-financial firms collected by the French central bank. The second source is AnaCredit France, a loan-level credit register of French banks, provided by the French central bank. I aggregate this second database at the firm-level and merge it with Fiben data. Using the resulting dataset, I show that rents paid to CRE investors represent a sizable and growing share of non-financial firms' value added. Moreover, CRE investors are structurally more leveraged, more likely to use their assets as collateral in lending relationships, and display a stronger correlation between investment and credit growth than other non-financial firms. Because their assets are more easily pledgeable, they are able to take on higher debt levels.

Second, building on this stylized fact, I construct a general equilibrium model with two production sectors. In the model, final good producers combine three inputs that are imperfect substitutes: *(i)* labor that they rent from households, *(ii)* standard capital that they accumulate subject to adjustment costs, and *(iii)* CRE that they rent from CRE investors. On the other side of this rental market,

[1]See, for instance, the European Systemic Risk Board (ESRB) recommendation to investigate the need for CRE-specific tools: recommendation n°2022/9.

CRE investors produce CRE by simply accumulating capital subject to adjustment cost. Both types of firms are owned by households and can finance capital accumulation through retained earnings and debt, but face financial frictions. First, because of agency problems between households and firms' managers, households discount dividends at a higher rate than interest payments. This implies that for firms, the cost of debt is lower than the cost of equity. Second, lenders impose a borrowing constraint in which firms' capital stock is used as collateral, as in Kiyotaki and Moore (1997). Crucially, I allow for sectoral heteroegenity in capital pledgeability. I then show analytically that capital pledgeability affects the expected marginal product of capital, and can thus lead to sectoral misallocation, both at the steady state and in face of credit supply shocks.

I then calibrate the model based on French macro and micro data, then assess the size of sectoral misallocation. In particular, I calibrate sectoral capital pledgeability such that it matches firm-level evidence: CRE investors are thus more leveraged and more exposed to credit supply shocks than final good producers. This simple difference in capital pledgeability generates sizable capital misallocation, as measured by spreads between capital-specific expected marginal products. At the steady state, the overall funding cost of CRE investors is lower because looser financial constraints allow these firms to finance a larger share of their capital accumulation with debt rather than with costly equity. This enables them to sustain a larger capital stock than final good producers, which yields lower expected return for this sector. In addition, in face of an exogenous increase in credit supply, this spread in expected returns further widens. I also simulate a qualitatively realistic credit cycle, defined as a sequence of small positive credit supply shocks, followed by a large one-off negative supply shock, as in Ferrero et al. (2024). I show that this exercise matches well historical evidence on sectoral heterogeneity in asset prices and credit allocation (Ivashina et al., 2024; Müller and Verner, 2024).

Because the two types of capital are imperfect substitutes, I finally show that this misallocation is suboptimal. In particular, I compute welfare gains of policy intervention correcting this misallocation, in the form of redistributive taxes on debt across production sectors. I find that taxing debt issued by CRE investors to subsidize debt issued by final good producers is welfare-improving, as it reallocates resources toward the most financially constrained sector, thereby increasing the consumption-to-labor ratio. Welfare gains are maximized for a tax equivalent to a 80-basis-point increase in the interest rate charged to CRE investors, which enables to fund a subsidy of around 40 basis points to final good producers. Considering different transition scenarios, I find that welfare gains are maximized if the policy is implemented during the boom phase of a credit cycle, as

it leans against the rise in misallocation and thus ensures a more efficient use of resources in the boom phase, on top of bringing the economy to a more desirable steady state.

These findings have tow important policy implications. First, they underline the importance of *sectoral* misallocation, while the literature has been mostly focused on *idiosyncratic* misallocation—see for instance David and Venkateswaran (2019). Although both phenomena can coexist and be equally important, policy implications are more straightforward for the former, as targeted firms are more easily identifiable. Second, these findings bring misallocation into the realm of macroprudential policies, which aim to "protect the economy from disruptions in the financial sector".[2] As the misallocation this paper documents stems from financial frictions and is amplified by credit cycles, this makes macroprudential authorities the most natural authorities to implement the type of policies discussed in this paper.

Literature review. This paper contributes to four strands of literature. It first contributes to the literature on capital misallocation by highlighting the role of credit frictions in generating *sectoral* misallocation. Resource misallocation has been extensively studied—see Hopenhayn (2014) and Restuccia and Rogerson (2017)— and has permeated the business-cycle literature after two decades of low investment and growth despite historically low interest rates (OECD, 2025). A first set of studies highlights how such an environment can create *intra-sectoral* misallocation, as credit expansion tilts production toward low-productivity firms (Aghion et al., 2019; Albuquerque, 2024; Asriyan et al., 2025; González et al., 2024a,b). My paper is closer to a second set of studies that focuses on *cross-sectoral* misallocation.[3] Bleck and Liu (2018) study a two-sector economy with heterogeneous financial frictions and show that excessive credit expansion can crowd out the most financial constrained sector. Within this strand, some papers focus on the specific role played by real estate. In particular, Hirano and Stiglitz (2025) show in a two-sector model how the design of credit policies can affect long-term growth by feeding land speculation at the expense of productive investment. The contribution of this paper is to study sectoral capital misallocation at business-cycle frequency and to highlight the role of heterogeneous capital pledgeability. Another key contribution is to reflect on the design of policy interventions correcting sectoral misallocation.

A second strand studies the role of CRE in shaping macroeconomic outcomes. The literature has emphasized the role of this asset in alleviating financial frictions faced by non-financial firms,

[2]See European Commission website, Macroprudential Policy.

[3]Bridging the gap between these two sets of studies, Li et al. (2025) notes that cross-sectoral differences in markup dispersions can be tied to sector-specific borrowing constraints.

especially smaller ones (Chaney et al., 2012; Gupta et al., 2021; Hericourt et al., 2024). In particular, Grjebine et al. (2023) highlight the role of booming real estate prices on aggregate productivity through reallocation effects, but focuses on non-financial firms outside the real estate and construction sectors. However, firms are also renters of real estate, which can weaken the link between CRE prices and investment at the *micro* level (Fougère et al., 2019). Considering the behavior of suppliers on the rental market is therefore central to assessing the strength of the link between CRE prices and investment at the *macro* level. Indeed, Duca and Ling (2020) argue that during the run-up to the GFC in the US, credit flowed disproportionately to CRE investors, who significantly contributed to the boom-bust episode. Similarly, Berg et al. (2023) find that in Germany, quantitative easing mostly benefited CRE investors over other non-financial firms. I rationalize these findings by emphasizing the looser borrowing conditions faced by CRE investors and by showing how these investors can affect macroeconomic outcomes.

This paper also contributes to the literature on the macroeconomic effects of financial shocks by introducing sectoral heterogeneity. The existing literature typically considers a representative corporate sector subject to collateral shocks (Jermann and Quadrini, 2012; Iacoviello, 2015; Becard and Gauthier, 2022). However, financial crises are generally associated with heterogeneous credit growth across sectors, with credit to CRE investors being particularly procyclical (Ivashina et al., 2024; Müller and Verner, 2024). This paper rationalizes these findings by incorporating CRE investors into a general equilibrium model with firm-level financial frictions. In doing so, I depart from collateral-shock frameworks, as the leverage of non-financial firms is empirically rather stable and heterogeneous across sectors. Instead, I focus on credit supply shocks, in line with the literature on residential real estate crises (Justiniano et al., 2019). I then show that CRE investors constitute an important transmission channel of these shocks to the real economy.

This paper finally contributes to the literature studying the effects of macroprudential policies, *i.e.*, policies aimed at mitigating the macroeconomic effects of financial frictions and shocks (Mendoza, 2010; Bianchi, 2011). Following the GFC, the literature has emphasized how these policies can be effectively applied to households to stabilize consumption in the face of residential real estate boom-bust cycles (Ferrero et al., 2024). Muñoz and Smets (2022) focus on the regulation of real estate investors—both in commercial and residential markets—as they can represent a leakage channel for standard residential real estate macroprudential regulations. The effectiveness of macroprudential policies for non-financial firms *per se*, by contrast, remains relatively understudied. Drechsel and Kim (2024) argue that the shape of the borrowing constraint firms face is

essential for macroprudential policy design. This paper instead emphasizes the role of sectoral heterogeneity among firms in shaping the design of macroprudential policies.

The rest of the paper is structured as follows. Section 2 presents stylized facts on CRE investors. Section 3 introduces a general equilibrium model featuring CRE investors and derives analytical results on sectoral misallocation. Section 4 studies sectoral capital allocation in a calibrated version of the model. Section 5 assesses the efficiency of redistributive sectoral taxes on credit. Section 6 discusses the main assumptions of the model and propose some robustness checks. Section 7 concludes.

2 Stylized facts on CRE investors

This section presents some stylized facts on CRE investors by combining the financial accounts of French non-financial firms and the corporate credit registers of French banks. I show that rents paid to CRE investors by other non-financial firms represent a sizable and growing share of their production costs. Moreover, CRE investors are structurally more leveraged, more likely to use their assets as collateral in lending relationships, and display a stronger correlation between investment and credit growth than other non-financial firms.

2.1 Data

I build a yearly firm-level dataset ranging from 1999 to 2023 by combining two data sources. The first source is Fiben (*Fichiers Bancaires des Entreprises*), a database gathering financial accounts, financial links, and credit ratings of a large sample of French firms collected by the French central bank. The sample covers firms with more than 750,000 euros of sales and represent approximately 70% the value added of French firms. I retrieve from this data source some identification variables: the national identification number (SIREN), age, 5-digit NACE sector code, and whether the firm is part of a corporate group. I also retrieve a number variables linked to firms' balance sheets and income statement: sales, rents paid, value added, Earning before Interests, Taxes, Depreciation and Amortization (EBITDA), net result, total assets, liquid assets, equity, total financial debt, financial debt with a residual maturity below one year. Finally, I also retrieve credit ratings produced by the French central bank. These credit ratings are based on accounting documents and more qualitative information, such as individual interviews conducted by analysts with firms' managers. These credit ratings feature 12 categories (21 since 2022) and are updated at least annually. They are available only since 2004 in the Fiben database. For each firm and year, I keep only the rating

which is the closest to the end of year, and merge it with my main dataset based on the firm identification number. Although consolidated accounts are also available in the database, I choose to focus on unconsolidated accounts as they cover a larger set of firms with homogeneous accounting definitions across time.

The second source is AnaCredit France (Analytical Credit Datasets), a credit register of French banks collected by the French central bank, and which is part of a broader Eurosystem database. This database records at a monthly frequency any credit instrument granted by a French bank to a moral person since 2019, for all credit relationships exceeding 25,000 euros. I retrieve from this dataset the outstanding amount of each instrument, and the type of collateral that is pledged by the borrower. I then compute at the firm level the mortgage ratio, which is the share of credit recorded in the register that is guaranteed by physical capital (*i.e.* commercial real estate, residential real estate, other physical capital). I only keep end-of-year observations to match the frequency of the Fiben data. I then merge on the resulting data on the Fiben dataset, keeping only firms appearing in the Fiben dataset.

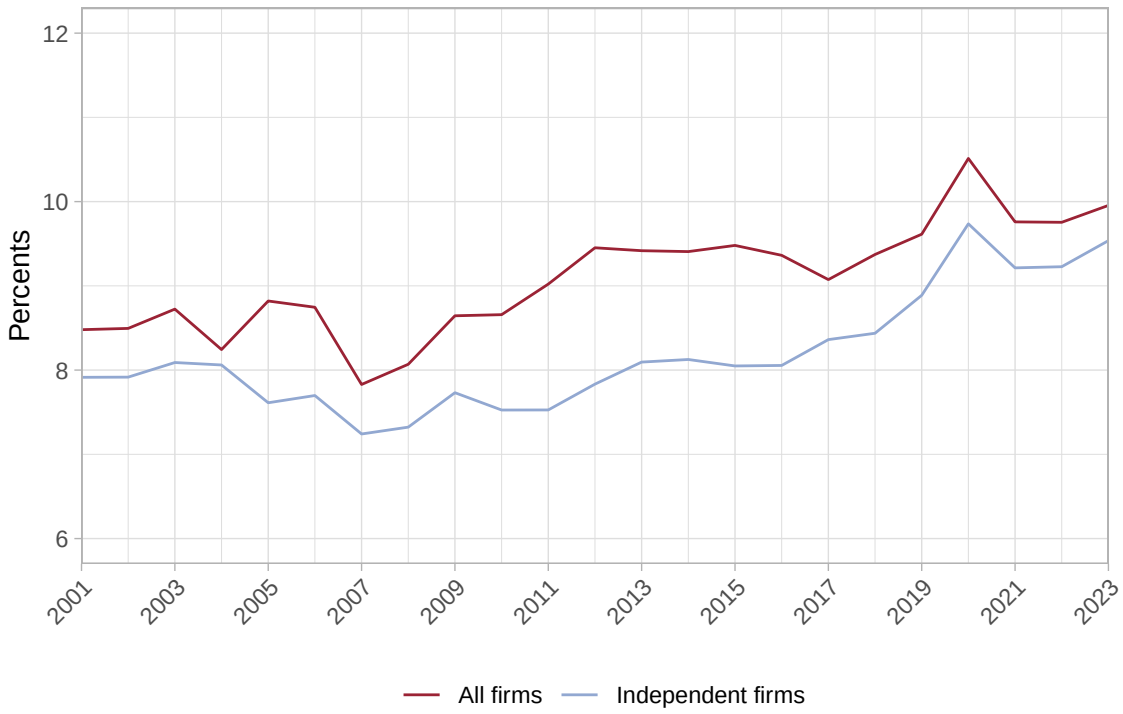
I then merge AnaCredit data on Fiben data, keeping only firms appearing in Fiben data. In this resulting dataset, I consider as CRE investors firms with NACE sector code 6820B: these are firms whose purpose is to lend non-residential properties. As the dataset only covers French non-financial firms, it leaves aside a number of active CRE investors on French markets, namely CRE investment funds and foreign CRE investors. Note, however, that these investors can own special-purpose vehicles located in France that are accounted for in the dataset, provided that they exceed the Fiben sales threshold. I transform the resulting dataset in several ways. I exclude from the sample the following sectors: financial services, public administration, and residential real estate investors. Note that construction firms are kept in the sample but not labeled as CRE investors. I keep only firms with strictly positive sales, non-financial fixed assets, equity and debt. I winsorize all variables at the top and bottom 1% of the distribution, except for age and size which I winsorize at the top 1%.

2.2 Fact 1: How large is the CRE sector?

Assessing the economic size of CRE investors could be done by computing the share of physical capital they own in my sample. However, as explained above, my dataset does not cover all CRE investors active on the French market, while it covers a very large share of value added. Put differently, the data provide a better coverage income flows than of capital stocks. A simpler

approach then consists in adopting a renter perspective, by computing rents paid by French firms (excluding CRE investors). This is the methodology adopted in Figure 2, which plots the ratio of rents to value added, which I define here as the difference between production and intermediary expenses except rents. The share of rents in valued added for the whole sample is sizable across the period, at least 8% of value added. Assuming that the cost of capital accounts for 35% of value added as it is usually done in quantitative macroeconomic models (Smets and Wouters, 2007), this means that a third of capital is in fact rented rather than owned. In addition, this value was stable until the GFC but rose progressively through the 2010s to reach 10% at the end of the period.

Figure 2: Rents over value added of French non-financial firms



Note: CRE investors are excluded from the sample. Value added is defined as the difference between production and intermediary expenses except rents. Independent firms are firms that are not part of group of non-financial firms..
Sources: Fiben.

Note however that all rents paid by these firms necessarily accrue to genuine CRE investors. In particular, it is possible that some firms pay these rents to a special purpose vehicle of the group which it is part of. This strategy would artificially increase the share of rents in value added, and would rather capture the complexity of a corporate groups rather than the economic weight of CRE investors. To alleviate this concern, I compute the ratio by restricting the sample to independent firms, *i.e.* firms that are not part of a group. This also means excluding the largest firms from the

sample. In any case, Figure 2 shows that the share of rents over value added decreases slightly but keeps the same magnitude and trend.

2.3 Fact 2: Are CRE investors more leveraged?

Are CRE investors structurally more leveraged than other non-financial firms? To investigate this, I estimate the following equation:

$$y_{it} = \alpha_t + \beta CRE_i + Z'_{it}\gamma + \varepsilon_{it} \tag{2.1}$$

y_{it} denotes across specifications leverage (defined as financial debt over total assets or as financial debt over financial debt plus equity) or the mortgage ratio. CRE_i is a dummy variable equal to one if the firm is a CRE investor, based on its sector code. Z'_{it} includes firm-level controls: age, log of total assets, operating return on assets (defined as earnings over lag total assets), net return on assets, liquidity (cash over sales), short-term debt (share of financial debt with a residual maturity below one year), credit rating. Finally, α_t is a time fixed effect which excludes cyclical variations to focus on structural cross sectoral heterogeneity. Standard errors are clustered at the firm and year levels.

Results are reported in Table 1. Controlling for years only, CRE investors have on average a 51.9% leverage (defined as debt over total assets), while that of other firms amount to 20.9% (column 1). Controlling for firms' characteristic, and notably their credit ratings, still yields a 22 percentage points difference between the two sectors (column 2). Defining leverage as debt over debt plus equity yield simuilar results (columns 3 and 4). This differential in leverage can be related to the share of mortgage in credit. In fact, 55% of CRE investors' credit is explicitly guaranteed by a physical asset, while this figure amount to only 16.1% for other firms (column 5). Including firm level controls still yield a 37.1 percentage points difference between the two sectors (column 6). These findings are robust to the exclusion of firms that are part of a non-financial groups (cf. Table 4 in Appendix A). Overall, these findings highlight the radical differences in funding structure between the two sectors. They suggest that this difference comes from the fact that the assets of CRE investors are more easily pledgeable, which enable them to take on more debt.

Table 1: Liability structure

	Dependent variable:					
	Debt / total assets		Debt / debt and equity		Mortgage / bank credit	
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.209***	0.106***	0.390***	0.227***	0.161***	0.157***
	(0.0003)	(0.004)	(0.0004)	(0.007)	(0.0004)	(0.021)
CRE investor	0.308***	0.219***	0.242***	0.150***	0.391***	0.371***
	(0.013)	(0.008)	(0.014)	(0.006)	(0.022)	(0.023)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm-level controls	No	Yes	No	Yes	No	Yes
Observations	4,884,312	3,786,807	4,884,312	3,786,807	1,035,250	934,817
R^2	0.038	0.213	0.019	0.261	0.016	0.025

Note: *p<0.1; **p<0.05; ***p<0.01
Clustered standard errors, in parenthesis, at the year and firm levels.

2.4 Fact 3: Is CRE investment more credit-sensitive?

Does this *cross-section* difference in funding structure imply a tighter link between credit and investment *across time* for CRE investors? To test this, I estimate the following equation:

$$\frac{\Delta K_{it}}{A_{it-1}} = \alpha_i + \alpha_{st} + \beta_0 \frac{\Delta B_{it}}{A_{it-1}} + \beta_1 \left(\frac{\Delta B_{it}}{A_{it-1}} \times CRE_i \right) + Z'_{it}\gamma + \varepsilon_{it} \tag{2.2}$$

ΔK_{it} denotes across specification the change in total or fixed assets (machines, real estate, patents, goodwills, etc.). A_{it} denotes total assets. ΔB_{it} is the change in financial debt. Z'_{it} are firm-level controls. As in the previous subsection I include operating return on assets, net return on assets, liquidity, short-term debt and credit rating. I add to this list changes in issued shares scaled by lag total assets, so as to control for firms' equity issuance. I further restrict the sample to firms with at least three reporting years. α_i is a firm fixed effect and α_{st} is a sector by time fixed effect. Despite these controls, this estimation is not to be interpreted in a causal sense, in particular because it is subject to reverse causality: coefficients measure the effect of a credit supply shock on investment, but also the reliance on credit to fund an investment opportunity.

Results are reported in Table 2. A one percentage point increase in credit is associated to a 0.66 percentage point increase in capital the same year for CRE investors, while it reaches only 0.44 percentage point for other firms (column 1). Including firm level-controls yields very similar results (column 2). The investment behavior of CRE investors seems thus more reliant on credit

access. This means that a rise in credit, controlling for equity issuance, does not increase one-to-one investment, as lenders may require borrowers to finance at least part of a project with inside funds. However, this constraint appears looser for CRE investors, who need less inside equity to fund their projects. This higher correlation between credit and investment translates into a tighter link between credit and growth for CRE investors. Indeed, a one percentage point increase in credit in period t is associated to a 0.96 percentage point increase in total assets in $t+1$ for non CRE investors. For CRE investors, this figures reaches 1.2 percentage point (columns 3 and 4). Interestingly, this value is virtually identical to one for non CRE investors while it is largely greater than one for CRE investors: credit is neutral for standards firms, while there is a positive multiplier of credit for CRE investors, potential linked to increased in asset valuations. These findings are robust to the exclusion of non-financial groups (cf. Table 5 in Appendix A).

Table 2: Credit growth and investment

	Dependent variable:			
	Fixed assets		Total assets in t+1	
	(1)	(2)	(3)	(4)
Credit	0.436***	0.416***	1.003***	0.965***
	(0.023)	(0.026)	(0.019)	(0.016)
Credit x CRE investor	0.222***	0.221***	0.221***	0.227***
	(0.022)	(0.023)	(0.031)	(0.031)
Sector x Year fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Firm controls	No	Yes	No	Yes
Observations	4,645,624	3,688,252	4,190,712	3,286,127
R^2	0.393	0.431	0.376	0.465

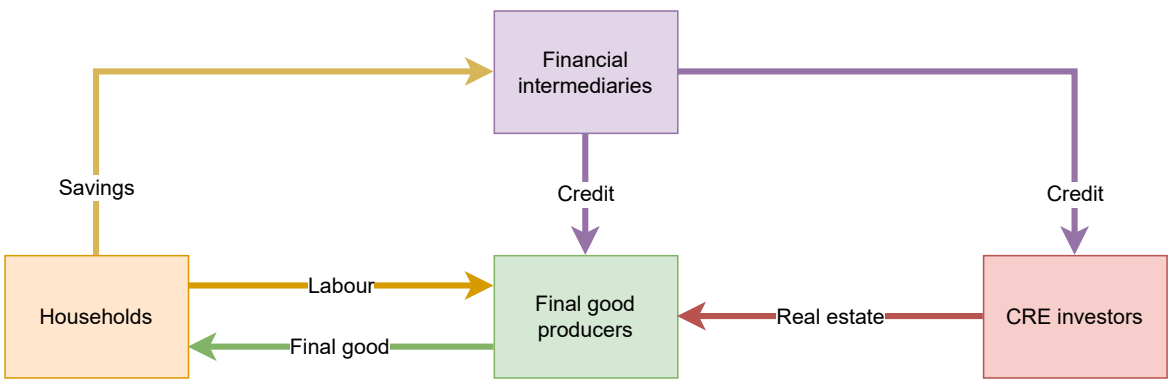
Note: *p<0.1; **p<0.05; ***p<0.01
Clustered standard errors, in parenthesis, at the year and firm levels.

These findings indicate that CRE investors play a significant economic role, and that they rely extensively on credit as their assets seem more frequently used as collateral. To study the macroeconomic implications of these facts, I build in the next section a dynamic stochastic general equilibrium model embedding this salient feature of CRE investors.

3 A macro-finance model with CRE investors

This section introduces a two-production-sector general equilibrium model. The model features final good producers, CRE investors and households. I also introduce simple pass-through financial intermediaries between households and the two production sector, in order to discuss the effect of exogenous credit supply shocks. Households supply labor to final good producers and funds to financial intermediaries. In turn, financial intermediaries supply credit to final good producers and CRE investors, who both accumulate capital. Figure 3 provides an overview of the model. I present the problem of each agent in turn before analyzing in more detail the effects of heterogeneous capital pledgeability on expected capital returns. Appendix B reports all equilibrium conditions, adding to the baseline model presented in this section the extensions discussed in section 6.

Figure 3: Model overview



3.1 Final good producers

The representative final good producer (denoted c) is fairly standard, except that she uses CRE in her production process. More precisely, the production technology takes the form of embedded Constant Elasticity of Substitution (CES) functions. The final good producer first combines a capital bundle X_{ct} with labor N_t, following a standard Cobb-Douglas technology with elasticity of substitution $\alpha \in (0, 1)$, such that output is:

$$Y_t = X_{ct}^{\alpha} N_t^{1-\alpha} \tag{3.1}$$

This capital bundle is itself produced by combining CRE H_t and standard capital K_{ct}, which are imperfect substitutes: the firm needs some specific capital that she owns (such as machines or

adapted production sites), but also standardized properties that she rents (such as offices, warehouses or shopping centers). More formally, these inputs are combined with elasticity of substitution $1/(1-\omega)$ where $\omega \in (-\infty, 1)$:

$$X_{ct} = \left(\nu H_t^{\omega} + (1-\nu) K_{ct-1}^{\omega} \right)^{\frac{1}{\omega}} \tag{3.2}$$

And with $\nu \in (0,1)$ the relative weight of CRE in the aggregate capital bundle.

Labor is rented by final good producers to households at wage w_t, while CRE is rented by final good producers to CRE investors at rent r_{ht}. Own-use capital is on the contrary directly held by the final good producer. The law of motion of this capital is:

$$K_{ct} = (1-\delta) K_{ct-1} + \left[1 - \frac{\varphi_c}{2} \left(\frac{I_{ct}}{I_{ct-1}} - 1 \right)^2 \right] I_{ct} \tag{3.3}$$

Where I_{ct} denotes investment, $\delta \in (0,1)$ is the depreciation rate of capital and $\varphi_c \in \mathbb{R}^+$ controls for the strength of investment adjustment costs for final good producers. The presence of this adjustment cost means that the market value and the book value of capital differ out of steady state. The resulting market value of capital Q_{ct} is then determined in equilibrium as the Lagrange multiplier associated to the law of motion of capital.

The final good producer can fund herself in two ways. She can first use internal resources by reducing dividend payments d_{ct}.[4] In addition, she can issue inter-period bonds B_{ct} with gross interest rates $1 + r_t$. Her borrowing capacity is however capped to a fraction of the assets she owns, because of a standard frictions on the credit market (Kiyotaki and Moore, 1997; Iacoviello, 2005), such that:

$$B_{ct} \leq \theta_c Q_{ct} K_{ct} \tag{3.4}$$

Where $\theta_c \in (0,1)$ is the loan-to-value imposed by financial intermediaries to the final good producer. The last constraint the final good producer faces is a budget constraint:

$$d_{ct} + (1 + r_{t-1}) B_{ct-1} + w_t N_t + r_{ht} H_t + I_{ct} \leq Y_t + B_{ct} \tag{3.5}$$

[4]Negative dividend then captures equity injection.

Given these constraints, the final good producer maximizes the flow of discounted dividends on behalf of households:

$$\mathcal{V}_{ct} = \mathbb{E}_t \sum_{j=0}^{\infty} \Lambda_{t,t+j} d_{ct+j} \tag{3.6}$$

Where $\Lambda_{t,t+j}$ is households' stochastic discount factor for dividends of period $t+j$ at period t, which the firm takes as given. The one-period ahead stochastic discount factor for dividends is thus $m_{t,t+1} = \mathbb{E}_t \frac{\Lambda_{t,t+1}}{\Lambda_{t,t}}$. I assume that the discount factor households apply to dividends is lower than the one they apply to interest payments, such that $\Lambda_{t,t+j} = \gamma^j \lambda_{t,t+j}$, with $\gamma \in (0,1)$ and $\lambda_{t,t+j}$ is households' stochastic discount factor for interest payments, defined below. This simple modeling capture standard agency problems between shareholders and firms' managers, which creates a preference of firms for debt over equity. I show below that this assumptions leads to binding borrowing constraint at the steady state.[5]

3.2 CRE investors

The representative CRE investor (denoted h) produces CRE out of the capital she owns K_h through a linear production function:

$$H_t = K_{ht-1} \tag{3.7}$$

The law of motion of this capital is then:

$$K_{ht} = (1-\delta)K_{ht-1} + \left[1 - \frac{\varphi_h}{2}\left(\frac{I_{ht}}{I_{ht-1}} - 1\right)^2\right] I_{ht} \tag{3.8}$$

Where I_{ht} denotes CRE investment and $\varphi_h \in \mathbb{R}^+$ controls for the strength of investment adjustment costs for CRE capital. The resulting market value of commercial real estate Q_{ht} is then determined in equilibrium as the Lagrange multiplier associated to the law of motion of capital.

Note that I assume that the depreciation of CRE capital K_{ht} is equal to that of standard capital K_{ct}, although the raw depreciation rate of buildings is arguably lower that of machines, software or computers (Iacoviello, 2015). This assumption is made for three reasons. First, higher depreciation rate of standard capital are usually partially compensated by higher tax rebates, such that the gap

[5] The macro-finance literature usually rather assumes a tax rebate on interest payment to break the Modigliani-Miller theorem—see Jermann and Quadrini (2012) for instance. My approach is closer to two-agent models which assume that agents differ by their discount factors, impatient households borrowing from patient households. This approach through discount rate heterogeneity implies that the root of the credit friction lays in structural parameters, rather that in the design of the tax system. As a consequence, the inefficiency it generates cannot be solved by the political authority by simply removing the distortional tax system. This is of particular interest for this paper, as it will investigate welfare-maximizing corporate taxes.

in net depreciation is arguably low. Second, the capital stock of final good producers K_{ct} also encompasses building, but whose particularity is to be less pledgeable because more specific to the firm's production process, but not necessarily to depreciates faster. Finally, this assumption is made for tractability to focus the analysis on a specific dimension of corporate heterogeneity, *i.e.*, capital pledgeability. This boils down to assume that there is a fiscal authority which pays a subsidy on capital that depreciates more quickly, financed through non distortive lump-sum taxes on households, such that the effective depreciation rate is equal across sectors.

As for the final good producers, the CRE investor can fund herself in two ways: she can use internal resources by reducing dividend payments d_{ht}, and issue inter-period bonds B_{ct} with gross interest rates $1 + r_t$. Her borrowing constraint then writes as follows:

$$B_{ht} \leq \theta_h Q_{ht} K_{ht} \tag{3.9}$$

Where θ_h is the loan-to-value imposed by financial intermediaries to the CRE investor. In line facts 2 and 3 in section 2, I assume that $\theta_h \in (\theta_c, 1)$. The last constraint the CRE investor faces faces is a budget constraint:

$$d_{ht} + (1 + r_{t-1})B_{ht-1} + I_{ht} \leq r_{ht}H_t + B_{ht} \tag{3.10}$$

Given these constraints, the CRE investor maximizes the flow of discounted dividends on behalf of households:

$$\mathcal{V}_{ht} = \mathbb{E}_t \sum_{j=0}^{\infty} \Lambda_{t,t+j} d_{ht+j} \tag{3.11}$$

Where d_{ht} denotes dividends and $\Lambda_{t,t+j}$ is the same stochastic discount factor than for final good producers. Similarly, there is a preference for debt over equity through a lower stochastic discount factor applied to dividends, which implies that the borrowing constraint is binding at the steady state.

3.3 Financial intermediaries

To discuss the effect of credit supply shock in the model, I introduce a simple pass-through financial intermediary that faces an exogenous and time-varying balance sheet constraint, following the recent macro-finance literature on residential real estate (Justiniano et al., 2019; Ferrero et al., 2024). The representative financial intermediary issues two instruments bought by households: deposits D_t (paid at rate r_{dt}) and securities S_t (paid at rate r_{st}). This financial intermediary grants

credit B_t indifferently to the final good producer and the CRE investor. Her balance sheet is thus the following:

$$B_t = S_t + D_t \tag{3.12}$$

In addition, she faces an exogenous and time-varying balance sheet constraint κ_t:

$$S_t \geq \kappa_t B_t \tag{3.13}$$

With $\kappa_t = \kappa + \zeta_{\kappa t}$, where $\zeta_{\kappa t}$ is an AR(1) process:

$$\zeta_{\kappa t} = \rho_\kappa \zeta_{\kappa t-1} + \varepsilon_{\kappa t} \tag{3.14}$$

Where $\varepsilon_{\kappa t}$ is an exogenous shock and $\rho_\kappa \in (0, 1)$ controls for the persistence of the shock. Because of portfolio adjustment cost at the household level introduced below, $r_{st} \geq r_{dt}$. It is thus less costly for financial intermediaries to collect deposits than to issue securities. The balance sheet constraint thus binds in equilibrium and directly affects credit spreads charged by financial intermediaries.

Given these constraints, the financial intermediary maximizes the following profit function:

$$\Pi_{bt} = r_t B_t - r_{st} S_t - r_{dt} D_t \tag{3.15}$$

Rewriting this function using the previous two constraints yields the following optimality condition:

$$r_t = (1 - \kappa_t) r_{dt} + \kappa_t r_{st} \tag{3.16}$$

Changes in the balance sheet constraint then capture credit supply shocks that move the spreads charged by intermediaries independently of borrowers' credit risk. This simple modeling choice deliberately abstracts from several features central to financial intermediation, including maturity transformation and interest rate risk. As a result, κ_t should be interpreted broadly: it captures any shift in interest rates that goes beyond (expected) movements in the short-term risk-free rate. This includes long-term interest rate risk (*i.e.*, the term premium), changes in intermediaries' risk aversion, or credit policies that affect lending costs without directly altering discount rates. The paper does not aim to provide a full evaluation of each of these shocks, but instead focuses on the transmission mechanism through which this class of shocks affects the real economy.

3.4 Households

The representative household consumes C_t units of the final good and supplies labor N_t to final good producers. She also collects dividends distributed by CRE investors and final good producers. In addition, she saves through two financial instruments: deposits D_t and securities S_t, both issued by financial intermediaries. Security holding is subject to portfolio adjustment costs, capturing their illiquid and risky nature, as in Ferrero et al. (2024). This adjustment cost is controlled by the parameter $\psi \in \mathbb{R}^+$. The household thus faces the following budget constraint:

$$C_t + D_t + S_t + \frac{\psi}{2}\left(\frac{S_t}{\bar{S}_t} - 1\right)^2 \bar{S}_t \leq w_t N_t + (1 + r_{dt-1})D_{t-1} + (1 + r_{st-1})S_{t-1} + d_{ht} + d_{ct} \tag{3.17}$$

Where $\bar{S}_t$ is the target level of securities that the household takes as given. For simplicity, I set $\bar{S}_t = \kappa(B_{ht} + B_{ct})$. As I will demonstrate below, this functional form implies that credit spreads moves if and only if there is a change in the balance sheet constraint of financial intermediaries—see also Ferrero et al. (2024) who adopt the same modeling technique.

The household derives utility from consumption, with risk-aversion $\sigma \in \mathbb{R}^+$. In addition, she derives disutility from labor, controlled by parameters $\chi \in \mathbb{R}^+$ and $\eta \in \mathbb{R}^+$:

$$U(C_t, N_t) = \frac{1}{1-\sigma} C_t^{1-\sigma} - \frac{\chi}{1+\eta} N_t^{1+\eta} \tag{3.18}$$

Each period, the household maximizes lifetime utility with discount rate $\beta \in (0, 1)$:

$$\mathcal{V}_t = \mathbb{E}_t \sum_{j=0} \beta^j U(C_{t+j}, N_{t+j}) \tag{3.19}$$

Hence the following stochastic discount factor for interest payments of period $t + j$ at period t:

$$\lambda_{t,t+j} = \beta^j \mathbb{E}_t C_{t+j}^{-\sigma} \tag{3.20}$$

3.5 Shocks and equilibrium

The only shock hitting this economy is the credit supply shock $\zeta_{\kappa t}$ introduced above. A competitive equilibrium in this economy is defined as a sequence of prices and quantities such that for a given sequence of prices, the first-order conditions of all agents and sectors are satisfied, and for a given sequence of quantities, prices adjust so that all markets clear. In particular, the credit market

clears:

$$B_t = B_{ct} + B_{ht} \tag{3.21}$$

And the resource constraint writes as follows:

$$Y_t = C_t + I_{ct} + I_{ht} + \frac{\psi}{2}\left(\frac{S_t}{\bar{S}_t} - 1\right)^2 \bar{S}_t \tag{3.22}$$

3.6 Heterogeneous capital pledgeability and expected returns

Final good producers and CRE investors interact directly on the rental market, and indirectly on the credit and equity markets, where they compete for funds. Crucially, these two sets of firms differ in one parameter: the pledgeability θ_c and θ_h of their respective capital stocks. In this subsection, I show analytically how cross-sector heterogeneity in capital pledgeability shape macroeconomic outcomes, at the steady state and in face of a credit supply shock.

I start by looking at the effect of heterogeneous capital pledegeability on capital returns at the steady state. I denote by r_c the marginal product of capital owned by final good producers. r_h is the CRE rental rate and corresponds in equilibrium to the marginal product of CRE. The capital demand of CRE investors is:

$$1 = m(1 - \delta + r_h) + \theta_h \mu_h \tag{3.23}$$

Where μ_h is the Lagrange multiplier associated to the borrowing constraint faced the CRE investor, which can be interpreted as the shadow cost of debt. Consider in parallel the credit demand of the CRE investor:

$$\mu_h = 1 - m(1 + r) = 1 - \gamma\beta \times \frac{1}{\beta} = 1 - \gamma \tag{3.24}$$

Because $\gamma < 1$, the cost of credit is lower than the cost of equity, so $\mu_h > 0$ and the borrowing constraint binds. We can thus rewrite capital demand:

$$1 - \delta + r_h = \frac{1 - (1-\gamma)\theta_h}{m} \Longrightarrow r_h = \frac{\gamma_h}{\beta} - 1 + \delta \tag{3.25}$$

Where $\gamma_h = \frac{1-(1-\gamma)\theta_h}{\gamma}$ is a sector-specific discount rate. It is a decreasing function of the share of capital that can be financed out of debt, as the cost of debt is lower than the cost of equity. Similarly for final good producers:

$$1 - \delta + r_c = \frac{1 - (1-\gamma)\theta_c}{m} \Longrightarrow r_c = \frac{\gamma_c}{\beta} - 1 + \delta \tag{3.26}$$

Where $\gamma_c = \frac{1-(1-\gamma)\theta_c}{\gamma}$. Because $\theta_h > \theta_c$, $\gamma_h < \gamma_c$, and the capital return spread is positive:

$$r_c - r_h = \frac{\gamma_c - \gamma_h}{\beta} > 0 \tag{3.27}$$

The required return on capital is lower for CRE investors: this is because their higher leverage enables them to benefit more from relatively low credit costs, compared to equity costs. This pushes CRE investors to accumulate more capital, up to the point where capital returns equate required returns. Because funding costs are unique across sectors (households investing in equity and financial intermediaries granting loans are *per se* indifferent between the two sectors), this creates capital misallocation: in an economy without financial friction (*i.e.*, equal credit and equity costs with $\gamma = 1$), capital pledgeability would be irrelevant and both sectoral required returns would be equal to $1/\beta - 1 + \delta$. In the model I study, because CRE investors are less constrained, it is more efficient to focus capital accumulation on this sector, leading to lower marginal productivity for CRE capital than for standard capital. The different leverage adjusted discount rates γ_h/β and γ_c/β thus lead to sectoral heterogeneity in expected returns.

I then investigate the dynamic properties of the log-linearized model around its deterministic steady state. Variables with a hat denote percent deviation from steady state, except for returns and Lagrange multipliers, which are in percentage point deviation from steady state. To do so, I first show in Appendix C that the credit spread is purely exogenous in the model, as in Ferrero et al. (2024). More formally, normalizing $\psi = \frac{1}{(1+r)\kappa}$, I show that $\hat{r}_t - \hat{r}_{dt} = \hat{\kappa}_t$. With this result in hand, I now turn to the effect of the shock on expected capital returns. Dropping the expectation operator for convenience, the sectoral demand for capital K_{it}, $i \in \{c, h\}$, writes:

$$Q_{it} = m_{t,t+1}\Big((1-\delta)Q_{it+1} + r_{it+1}\Big) + \theta_i \mu_{it} Q_{it} \tag{3.28}$$

Consider in parallel credit demand:

$$\mu_{it} = 1 - m_{t,t+1}(1 + r_t) \tag{3.29}$$

Combining the two:

$$\underbrace{(1-\theta_i)Q_{it}}_{\text{Cost}} = \underbrace{m_{t,t+1}\Big((1-\delta)Q_{it+1} + r_{it+1} - \theta_i(1+r_t)Q_{it}\Big)}_{\text{Benefit}} \tag{3.30}$$

The cost of investment today is diminished by the amount of debt the investor can take on. The benefit next period is equal to underappreciated capital at market value, plus its marginal product, minus the share of the investment financed through debt that has to be repaid. Log-linearizing and using the fact that $\hat{r}_t - \hat{r}_{dt} = \hat{\kappa}_t$:

$$\hat{Q}_{it} = \hat{m}_{t,t+1} + \frac{1}{1-\delta+r_i-\theta_i(1+r)}\left((1-\delta)\hat{Q}_{it+1} + \hat{r}_{it+1} - \theta_i\hat{r}_{dt} - \theta_i\hat{\kappa}_t\right) \tag{3.31}$$

As $\hat{r}_{dt} = -(1+r)\hat{m}_{t,t+1}$ and $1-\delta+r_i = \gamma_i/\beta$, re-arranging this expression yields:

$$\hat{r}_{it+1} = \theta_i\hat{\kappa}_t + \frac{\gamma_i}{\beta}(\hat{Q}_{it} - \hat{m}_{t,t+1}) - (1-\delta)\hat{Q}_{it+1} \tag{3.32}$$

To see more clearly the direct role played by capital pledgeability in affecting expected capital returns, one can take the partial derivative of the semi-elasticity of capital returns to credit supply:

$$\frac{\partial(\hat{r}_{it+1}/\hat{\kappa}_t)}{\partial\theta_i} = \underbrace{1}_{\text{Direct exposure}} - \frac{1-\gamma}{\beta\gamma}\left(\underbrace{\frac{\hat{Q}_{it}}{\hat{\kappa}_t}}_{\text{Financial accelerator}} - \underbrace{\frac{\hat{m}_{t,t+1}}{\hat{\kappa}_t}}_{\text{Dividend discount factor}}\right) \tag{3.33}$$

Because $\frac{\partial\gamma_i}{\partial\theta_i} = -\frac{1-\gamma}{\gamma}$.

This expression shows that capital pledgeability directly affects the semi-elasticity of expected returns to credit supply through credit exposure: a one percentage point increase in pledgeability leads to a one percentage point higher semi-elasticity. Through this channel, in case of an increase in credit supply (*i.e*, $\hat{\kappa}_t < 0$), expected returns drop by more for CRE investors than for final good producers, as $\theta_h > \theta_c$. Lower credit cost then pushes CRE investors to increase more their capital demand to match their lower expected returns, as long as marginal returns are a decreasing function of the capital stock.

This direct role of θ_i is complemented by two indirect channels. The first one is through capital price, and can be linked to the well-known financial accelerator (Bernanke et al., 1999). Because of investment adjustment costs, the relative price of investment varies over the cycle. As it directly enters the borrowing constraint, θ_i influences its ability to influence capital accumulation. As seen previously, the direct channels means that an increase in credit supply decreases expected returns and thus increase capital demand, under fairly general conditions. In the presence of investment adjustment cost, this increases the relative capital price, such that it can be reasonably assumed that $\hat{Q}_{it}/\hat{\kappa}_t < 0$. This implies that a higher θ_i further increases capital returns' semi-elasticity to

credit supply: a high capital pledgeability allows a firm to more easily absorb the increase in capital prices through debt.

The second indirect channel goes through dividends' discount factor. Indeed, as κ_t shifts capital demands, this in turn affects households' consumption and their supply of funds, thereby moving $\hat{m}_{t,t+1} = \sigma(\hat{c}_t - \hat{c}_{t+1})$ and spilling back to capital demand, as it moves all financing costs, both credit and equity. Determining the sign of $\hat{m}_{t,t+1}/\hat{\kappa}_t$ is however less straightforward here. As I model a closed economy and that the shock does not yield any direct productivity gain, it is reasonable to assume that consumption will drop on impact so that investment can rise, before increasing back to steady state, yielding a decrease in the stochastic discount factor and an increase in the real interest rate.[6] Therefore, $\hat{m}_t/\hat{\kappa}_t > 0$, and through this channel, a higher capital pledge-ability dampens the elasticity of capital returns to credit supply. This is because high-pledgeability imply lower exposure to the equity market, which is itself more exposed to shift in dividend discount factor than the credit market. Indeed, the expected return on dividend can be written as $\hat{r}_{mt} = \frac{1}{\gamma\beta}\hat{m}_{t,t+1}$, while $r_{dt} = \frac{1}{\beta}\hat{m}_{t,t+1}$. As long as $\gamma < 1$, $\frac{\partial \hat{r}_{mt}}{\partial m_{t,t+1}} > \frac{\partial \hat{r}_{dt}}{\partial m_{t,t+1}}$.

Overall, the total size of the indirect effects depends on the difference between the size of the financial accelerator channel and that of the dividend discount factor channel. Note that these indirect effect are mitigated by the size of the financial friction, as measure by γ. As γ gets closer to one, the roe of these indirect effect vanishes. More generally, the role of capital pledgeability on expected returns is conditional on a number of parameters and functional forms: the shape of the production function is crucial the response of capital demand, that of investment adjustment cost is central for the financial accelerator mechanism, etc. The next section will tackle this question by adopting a more quantitative approach, with the whole set of parameters being chosen so as to match micro and macro data.

4 Capital allocation along the credit cycle

4.1 Calibration

The period of the model is a quarter. The discount factor β targets a standard 2% risk-free rate, while γ targets a 5% equity premium, following Damodaran (2025).[7] The intertemporal elasticity

[6] In the numerical application next section, I will show that this dynamic is in fact a bit more complex in the presence of investment adjustment cost: although the discount rate does indeed fall in the medium run, it initially rises in the short run.

[7] However, as I linearize the model around the deterministic steady state, it only captures the part of the equity premium that is friction-related, not risk-related. Ilut et al. (2025) estimates that around 40% of this premium can be

of substitution σ is set to 2, and the inverse elasticity of labor supply η to 1. The capital share α and capital depreciation δ follow Smets and Wouters (2007). For convenience, I choose a Cobb-Douglas production for the capital bundle X_{ct}, by setting the elasticity of substitution ω between CRE capital and standard capital to 0. I however conduct sensitivity analysis with respect to this parameter which is hard to calibrate. Crucially, some key parameters are chosen in line with the the empirical exploration conducted in section 2. The share of CRE in output ν directly determines the rent to output ratio if $\omega = 0$, and given $\alpha = 0.35$, one has to set $\nu = 0.25$, such that $r_h H/Y \approx 9\%$. In the model, the capital pledgeability parameters θ_c and θ_h pin down the amount of capital that can be financed out of debt. Empirical evidence (facts 2 and 3 in section 2) suggest that 40% of capital is financed out of debt for final good producers, while this share rise to 65% for CRE investors.

To pin a value on investment adjustment adjustment costs φ_h and φ_c, I estimate the model based on four French data series from 2000:Q1 to 2025:Q3: real GDP, investment in real estate and construction goods by non-financial corporations, other investment by non-financial corporations, and a measure of credit spread. Credit spread are measured by taking corporate bond yields of European firms with a maturity between 5 and 7 years, minus expected policy rates over 5 years as reported by Krippner (2013). The resulting series is plotted in Figure 10 in Appendix A, along with sectoral investment growth rates. In line with subsection 3.6, both investment growth rates seem negatively correlated with credit spread, especially for real estate investment. As the model is a stationary and expressed in real terms, I perform some additional transformations so that the observables fit the model's structure: nominal GDP and investment are divided by the GDP deflator, in log-differences, and demeaned. On top of the credit spread shock $\varepsilon_{\kappa t}$ presented in details in the previous section, I introduce three more standard shocks: sectoral investment shocks ε_{ht} and ε_{ct} following Justiniano et al. (2011), as well as a labor productivity shock ε_{at}. All of these shocks are normally distributed with zero mean and standard deviation σ_κ, σ_h, σ_c and σ_a. They are autocorrelated of order one, with coefficient ρ_κ, ρ_h, ρ_c and ρ_a. These shocks are explicitly integrated in the full model in Appendix B. As the credit spread is exogenous in the model, it is directly pinned down by its empirical counterpart. The credit spread shock then captures slow-moving investment dynamics that are common to both non-financial sectors and correlated with credit growth. Sectoral investment shocks then capture more sectoral dynamics, as well as financial shocks unrelated to credit spreads.

attributed to frictions rather than aggregate uncertainty, suggesting another reasonable calibration would target a 2% equity premium.

The estimation follows a Bayesian approach applied to a first-order log-linearized version of the model, using the Kalman filter and a Monte Carlo Markov Chain with 500,000 draws. Computations are performed with the Dynare toolbox (Adjemian et al., 2024). Prior and posterior distributions are reported in Table 6 in Appendix A. Both investment adjustment costs turn to be relatively low, more than twice smaller than the prior, which is similar to Smets and Wouters (2007). One reason maybe that my credit spread shock capture slow-moving investment dynamics and investment shocks capture more short-run dynamics, while these studies only use the latter in their estimations, which leads to high adjustment cost to rationalize data. Another striking result is that the CRE investment adjustment cost φ_h is significantly lower than the standard investment adjustment cost φ_c. This can seem surprising, as it is no uncommon in the macroeconomics literature to assume that real estate is in fixed supply (Iacoviello, 2015). This assumption does not well apply to commercial real estate, which proved to be less rigid than other capital types. Indeed, one one hand, CRE capital can be regarded as more costly to scale up, as buildings are less divisible than machines, and require land, which is in fixed supply. However, this applies less to CRE than to residential real estate, as CRE can more easily built in less dense zones. In addition, note that investment adjustment costs also capture the ability to integrate new investment to the existing capital stock. In this regard, it is more difficult for a firm to integrate machines or software to its production process and thus its effective capital stocks, while the addition of a new building in the portfolio of a CRE investor does not require such effort. In addition, CRE can be built in less dense zones, while there seem to be a more pronounced demand for central zones on the residential real estate market.

Finally, up to to the first order, the balance sheet constraint of financial intermediaries κ and households' portfolio adjustment cost ψ only influence the dynamics of spreads in the model, and are thus only scaling factors of credit supply shocks. Chosen parameters are summarized in Table 3. All simulations presented below are based on first-order approximation around the deterministic steady state using the Dynare toolbox (Adjemian et al., 2024).

4.2 Credit supply shock and sectoral capital allocation

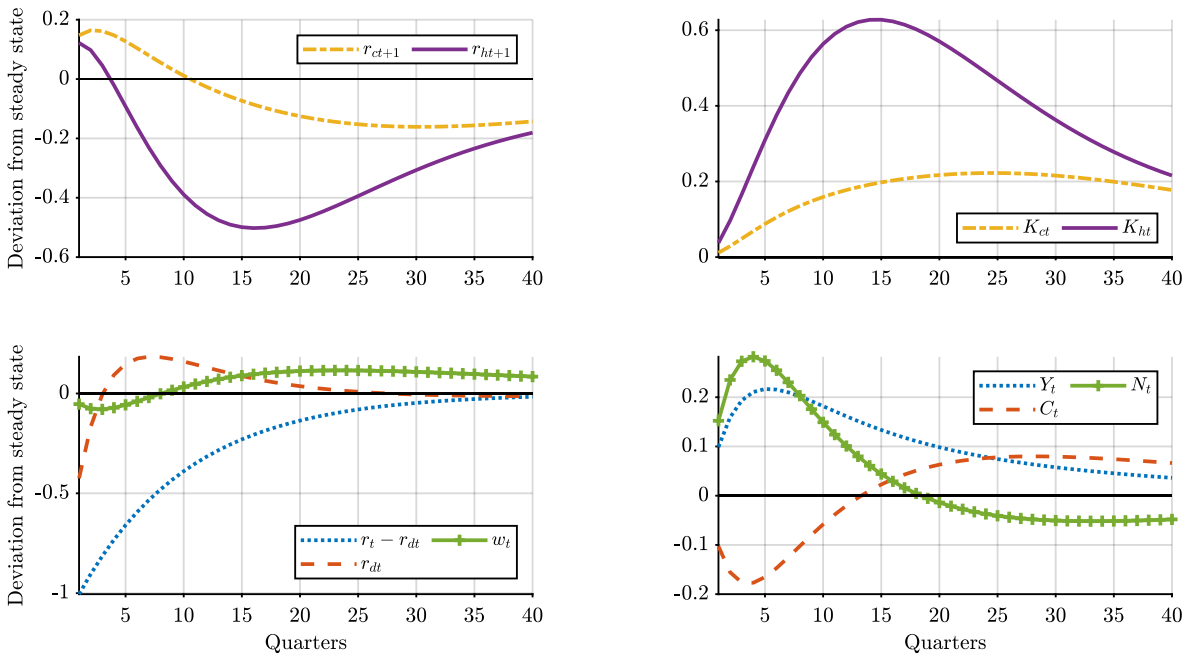
In this subsection, I study the response of the model to a one percentage point decrease in annualized interest rate spread, with autocorrelation of order one set to $\rho = 0.9$, in line with estimation results (see Table 6 in Appendix A). Figure 4 reports the response of the main variables to this shock. All variables are expressed in percent deviation from steady state, except interest rates

Table 3: Parameters

Description	Parameter	Value	Target/source
Deposit discount factor	β	0.995	2% yearly interest rate
Dividend discount factor	γ	0.988	5% premium (Damodaran, 2025)
Intertemporal substitution	σ	2	Standard
Inverse elasticity of labor supply	η	1	Standard
Capital share of output	α	0.35	Smets and Wouters (2007)
Capital depreciation	δ	0.025	Smets and Wouters (2007)
CRE and standard capital substitution	ω	0	Cobb-Douglas production
CRE share of capital	ν	0.25	$r_h H/Y = 9\%$ (section 2)
Leverage of final good producers	θ_c	0.4	$B_c/K_c = 40\%$ (section 2)
Leverage of CRE investors	θ_h	0.65	$B_h/K_h = 65\%$ (section 2)
Standard capital adjustment cost	φ_c	2.340	Estimation result (Table 6)
CRE capital adjustment cost	φ_h	1.315	Estimation result (Table 6)
Labor disutility	χ	1	Scaling factor
Financial intermediaries' constraint	κ	0.15	Scaling factor
Portfolio adjustment cost	ψ	0.2	Scaling factor

which are in annualized percentage point deviation from steady state.

Figure 4: Impulse responses to an increase in credit supply



As expected, the decrease in spread (blue dotted line, bottom left panel) leads to a progressive rise in the capital stocks of both sectors (upper right panel). However, the peak response of capital accumulated by CRE investors is three times larger than that of final good producers, leading

to a more pronounced fall in expected returns (upper left panel). Overall, this fits the analytical discussion in the previous section. However, this unbalanced capital accumulation takes time to unfold, after nearly a year: in the short run, both capital marginal productivity rise, while that of labor decreases (bottom left panel, green crossed line). This is because in the short run firms do not immediately scale up their capital stocks, and therefore increase their labor demand in the meantime.[8] Note that the real interest rate on deposits decreases (bottom left panel, red dashed line): households' fund supply increase more than firms' demand, as investment adjustment costs hinder capital accumulation in the short run. After a few quarters however, the interest rate starts to increase, as investment speeds up. This comes with a decrease in consumption, before increasing as capital accumulates: in the closed economy set-up I study, as there is no direct productivity gains from these shocks, the increase in investment can only be funded by a decrease in consumption. Consumption needs to fall to fund capital accumulation.[9] Finally, output follows a hump shape, driven by labor in the short run, and then by capital accumulation in the medium run, once investment adjustment costs are absorbed (lower right panel, blue dotted line).

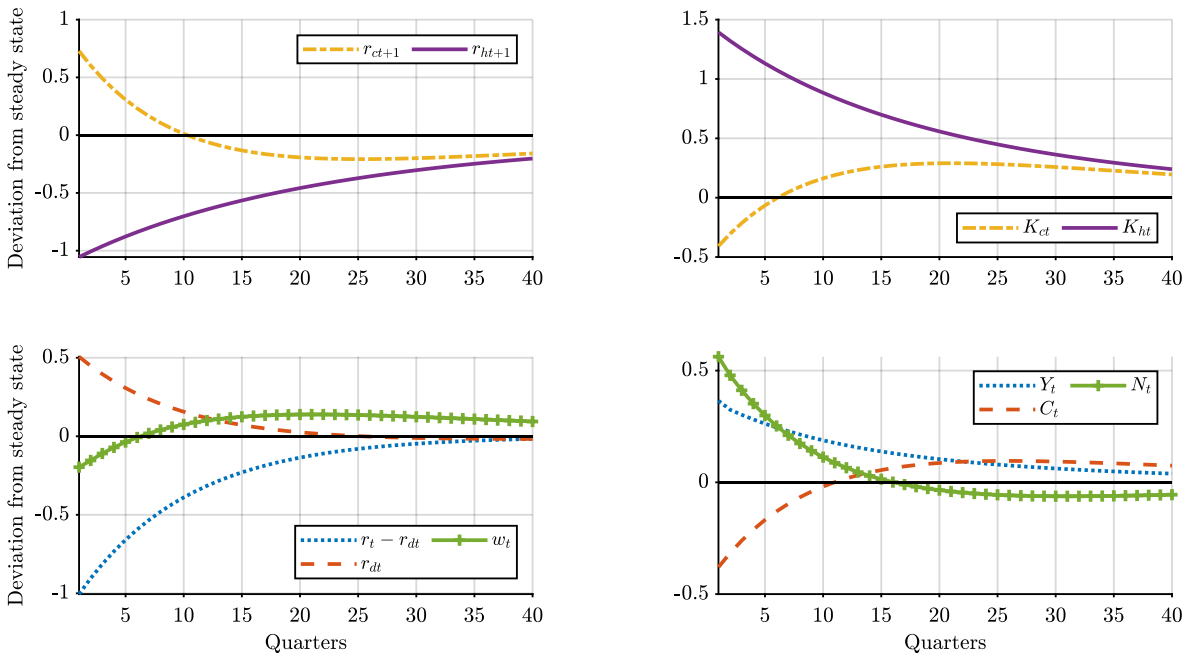
Investment adjustment costs reduce the gap in capital accumulation across sectors. Figure 5 plots the same shock, assuming $\varphi_c = \varphi_h = 0$. The heterogeneous capital accumulation rates becomes wider, and interestingly K_{ct} *decreases* in the short run. This is because in the absence of investment adjustment costs, the interest rate on deposits rise sharply, by 0.5 percentage points.[10] As discussed in the previous section, final good producers are more sensitive than CRE investors to the real interest rate on deposits, which then offset the decrease in credit costs. This suggests that CRE investment crowd out investment by final good producers: as CRE capital demand elasticity to credit spreads is higher, their increase in capital demand pushes up *all* interest rates, spilling over final good producers which are particularly sensitive to deposit interest rates. Because there is only one interest rate but heterogeneous capital demand elasticity, this creates potential for misallocation across sectors, as one crowds out the other.

[8] Because I assume separable utility, labor supply also increases, through a wealth effect linked to the low interest rate on deposits.

[9] This is by opposition to a more standard modeling of financial shocks as shocks to the marginal efficiency of investment (Justiniano et al., 2011), along with with no complementarity between hours worked and consumption households' welfare (Furlanetto and Seneca, 2014). However, the modeling of financial shocks as credit supply shocks, following Justiniano et al. (2019) and Ferrero et al. (2024), offers a more intuitive interpretation and is more suited to the study of a common financial shock hitting heterogeneous firms.

[10] The link between investment adjustment cost and the real interest rate is a standard result of the medium-scale DSGE literature (Christiano et al., 2005). These adjustment costs disconnect the real interest rate from the marginal product of capital: the presence of adjustment cost decreases capital demand, leading to a decrease of the real interest rate in the short run.

Figure 5: Impulse responses to an increase in credit supply - no investment adjustment costs



Note that the shock concerns only the ability of the banking sector to supply *intertemporal credit to non-financial firms*. As such, it is does not fully capture the effect of a realistic credit supply shock, that can also apply to *intra* temporal credit (typically, to finance working capital) or household credit. Compared to a credit crisis, it affects mostly the medium term behavior of the economy through investment dynamics. As my purpose is to rationalize an observed pattern of sectoral corporate heterogeneity, rather than provide an all-encompassing model fully replicating the effect of a credit shock, I do not judge this behavior problematic.

Overall, these findings confirm the intuition of the previous section and provide a more quantitative assessment: credit supply shock disproportionately affect CRE investors, leading to diverging capital accumulation processes between CRE investors and final good producers. Heterogeneous capital demand elasticities play a key role, notably by generating crowding out on financial markets.

4.3 Credit cycle and sectoral capital allocation

The previous subsection focused on the effect of a positive credit supply shock starting from the steady state. However, credit supply shocks do not usually occur in isolation: they are usually part of a credit cycle, defined as a sequence of small unanticipated positive credit supply shocks, fol-

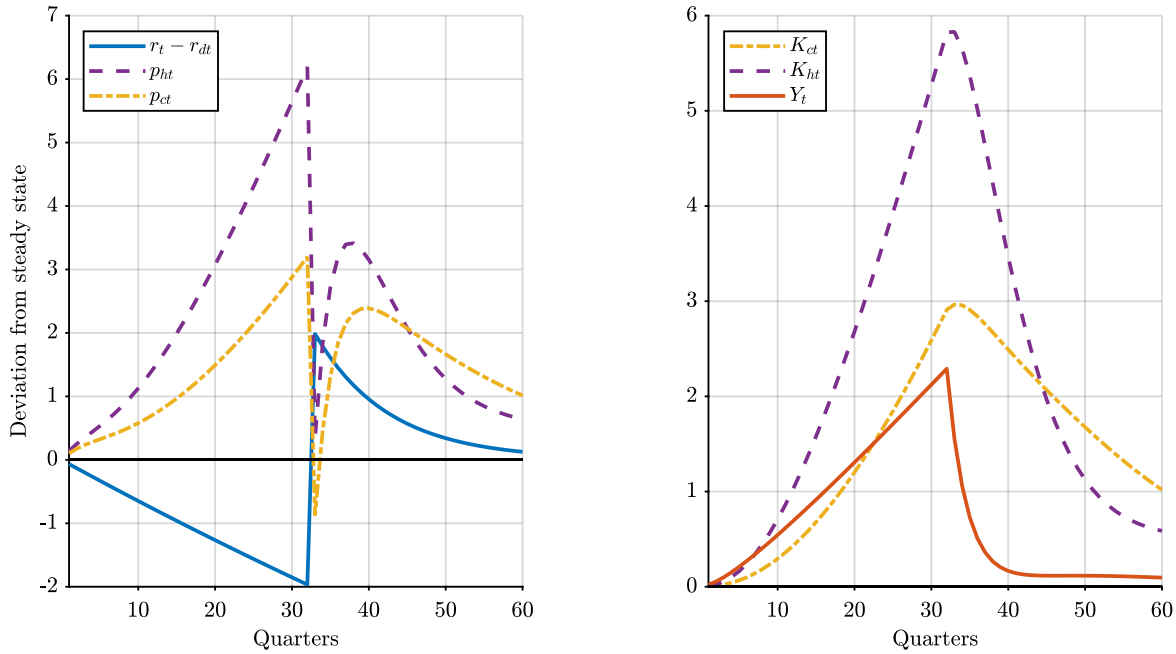
lower by a large unanticipated negative credit supply shocks that more than reverses the previous dynamic (Krishnamurthy and Muir, 2025). Following Ferrero et al. (2024), I calibrate a stochastic process that mimics a boom-bust cycle such as the GFC, through two shocks hitting κ_t: a nearly permanent component with autocorrelation set at 0.995, and a temporary component with autocorrelation set at 0.9. The boom phase is the result of a nearly permanent drop in spread for 32 quarters, which reaches -2 percentage points. The bust phase is the result of a nearly permanent shock that cancels the previous phase, completed by a temporary rise in spread of +2 percentage points. There are no other shocks afterwards. As the model is linear, this cycle is merely a linear combination of IRFs. Key insights from the previous subsection then remain, but this more realistic process enables to assess the ability of the model to fit some stylized facts on credit cycles and the role of CRE investors. In addition, the next section will consider the implementations of some public policies along this cycle.

Figure 6 reports the effects of such a cycle, the left panel presenting the interest rate spread in blue, along with the stock valuation of each sector, defined as $p_{st} = \mathbb{E}_t m_t(p_{st+1} + d_{st+1})$, $\forall s \in \{c, h\}$. Consistently with the empirical evidence reported in the introduction, the decrease in spreads has a stronger impact on the valuation of CRE investors than of other non-financial firms, nearly twice as large. When the crisis hits, both valuations immediately adjust downward, but then swiftly rise again, especially for CRE investors. This is because in the new high credit cost environment, equity financing is privileged to sustain the capital stock they previously accumulated. This rise in the capital stock is illustrated in the right panel: capital accumulation by CRE investors is nearly twice larger than that of final good producers. During the expansion phase, capital returns thus end up falling by more for CRE capital than for standard capital, suggesting inefficient resource allocation. This over-accumulation persists for some time after the crisis has hit because of investment adjustment costs. Despite this, the adjustment is relatively quick and even turns in favor of final good producers, as CRE investors are more sensitive to high credit costs.

5 Policy for sectoral misallocation

The previous section highlighted in a calibrated model that heterogeneous capital pledgeability structurally creates sectoral dispersion in expected capital returns. In addition, this dispersion is time-varying, as the cost of credit varies with credit supply shocks. The resulting hetereogeneity in capital accumulation is to some extent efficient, as capital accumulation is concentrated in firms which are the least exposed to financial frictions, *i.e.*, CRE investors. However, this excess accumu-

Figure 6: Simulated credit cycle



lation of CRE stumbles over imperfect substituability between capital types: unbalanced capital accumulation reduces production efficiency. This potentially leaves some room for a policy intervention reallocating resources across sectors, so as to increase production efficiency, and ultimately households' welfare.

5.1 Policy instruments and welfare gains

One fundamental friction in the model is the implicit agency problem between firms and household, which gives rise to a higher discount rate for dividends than for interest payments, creating a preference for debt for firms. CRE investors are less concerned than final good producers by this issue, as their higher capital pledgeability enables to partially absorb this financial friction. This implies that there is an under-provision of capital compared to the frictionless economy, but less so for CRE investors than for final good producers. A first best policy ensuring balanced budget would take the form of heterogeneous subsidies to capital accumulation across the two sectors, funded by lump-sum taxes on households.

This type of taxation is however a political challenge to implement. I instead assume that policymakers have access to proportional taxes and subsidies on firms' debt, denoted τ_{it}, $i \in \{c, h\}$. The fact that the tax is proportional to *debt* is a welcomed feature given that the heterogeneity in capi-

tal allocation I document stems from credit friction. The introduction of these taxes in the model introduced above changes only firms' credit demand, which can be re-written as:

$$1 + \tau_{it} = \mu_{it} + \mathbb{E}_t m_{t,t+1}(1 + r_t) \quad \forall i \in \{c, h\} \tag{5.1}$$

Note that for a tax level to be feasible, it must be that:

$$\tau_{it} \geq \mathbb{E}_t m_{t,t+1}(1 + r_t) - 1 \quad \forall i \in \{c, h\} \tag{5.2}$$

These inequality constraints stem form the fact that the policymaker moves the Lagrange multipliers on debt through the taxes she sets. Therefore, a feasible policy needs to respect $\mu_{it} \geq 0$, $i \in \{c, h\}$, yielding the above conditions. Moreover, I assume that the resulting policy is purely redistributive, *i.e.*, it must respect balanced budget:

$$\tau_{ht} B_{ht} + \tau_{ct} B_{ct} = 0 \tag{5.3}$$

This means that by setting τ_{ht}, τ_{ct} endogenously adjusts, so as to ensure balanced budget. Although this policy might appear as unorthodox from the perspective of a *fiscal* authority, it fits better in the toolkit of *macroprudential* authorities. In the European Union, these authorities have the ability to indirectly change the interest charged by banks to different types of borrowers through sectoral prudential requirements.[11]

To assess the efficiency of the policy, I compute welfare gains in consumption-equivalent terms, taking into account transition dynamics. This can be interpreted as the level of consumption the household in the no-policy economy would need to receive to be indifferent between the no-policy economy and the alternative policy considered. A positive value indicates that the policy is desirable. As there is only one representative agent, computations are naturally based on households' lifetime utility. Formally, expected welfare gains of a given policy ι_τ are defined as follows:

$$\sum_{t=0}^{\infty} \beta^t U\Big((1 + \iota_\tau) C_t, N_t\Big) = \sum_{t=0}^{\infty} \beta^t U(C_{\tau t}, N_{\tau t}) \tag{5.4}$$

[11] See ESRB website for sector specific risk weights or sectoral capital requirements. Note that because in the model financial intermediaries are simply pass-though vehicles, it is equivalent to directly tax firms or to impose sectoral capital requirements on banks. Relaxing this assumption and studying its implication for the practical design of the policy intervention I propose is left to further research.

Where $C_{\tau t}$ and $N_{\tau t}$ denote consumption and labor under a specific tax scenario. Using the definition of the utility function, this can be re-written as:

$$\iota_\tau = \left(\frac{\sum_{t=0}^{\infty} \beta^t U(C_{\tau t}, N_{\tau t}) + \sum_{t=0}^{\infty} \beta^t \frac{\chi}{1+\eta} N_t^{1+\eta}}{\sum_{t=0}^{\infty} \beta^t \frac{1}{1-\sigma} C_t^{1-\sigma}} \right)^{\frac{1}{1-\sigma}} - 1 \tag{5.5}$$

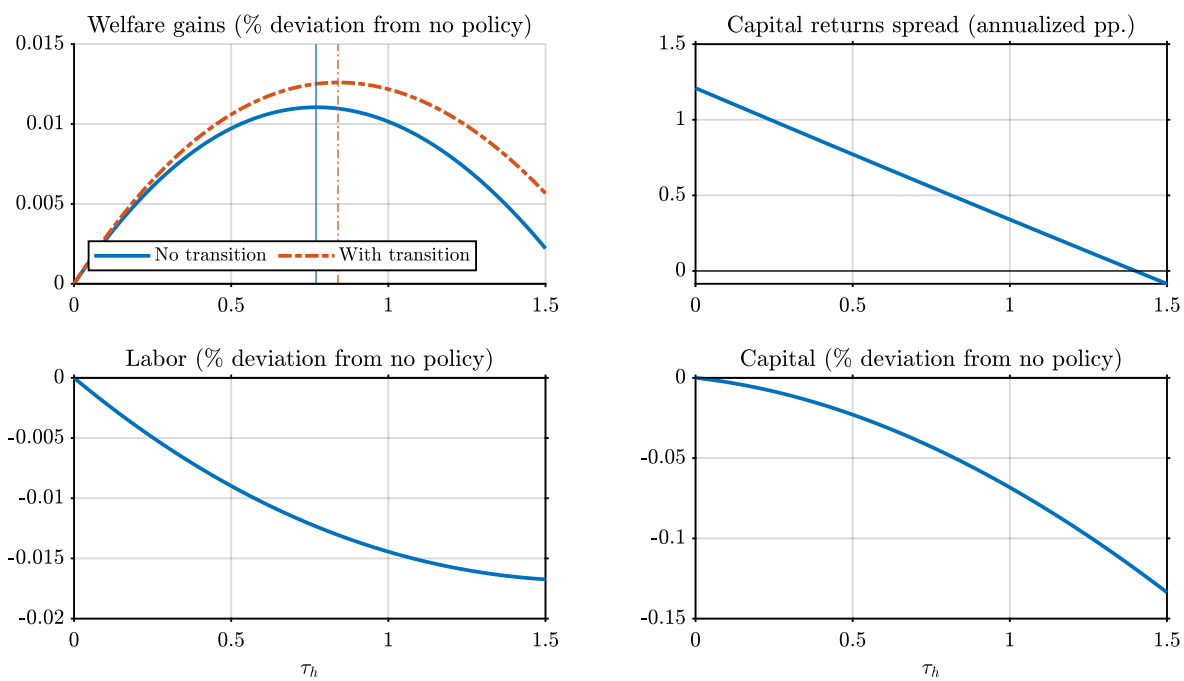
Although one policy can be optimal from an ex-ante perspective at $t = 0$, its political implementation may also depend on the *timing* of welfare gains. To asset those, I also compute period-by-period welfare gains $\iota_{\tau t}$ as follows:

$$\iota_{\tau t} = \left(\frac{U(C_{\tau t}, N_{\tau t}) + \frac{\chi}{1+\eta} N_t^{1+\eta}}{\frac{1}{1-\sigma} C_t^{1-\sigma}} \right)^{\frac{1}{1-\sigma}} - 1 \tag{5.6}$$

5.2 Constant tax rates

In this subsection, I investigate the welfare gains of a constant tax on CRE debt τ_h. I consider two cases: *(i)* welfare gains ignoring transition dynamics, and *(ii)* welfare gains taking into account transition dynamics. I compute welfare gains over an horizon of 1,000 quarters. I check that the system has converged to the new steady state at this horizon, and that the Lagrange multiplier on debt remains strictly positive. Figure 7 plots the results of these experiments.

Figure 7: Welfare gains of constant policy



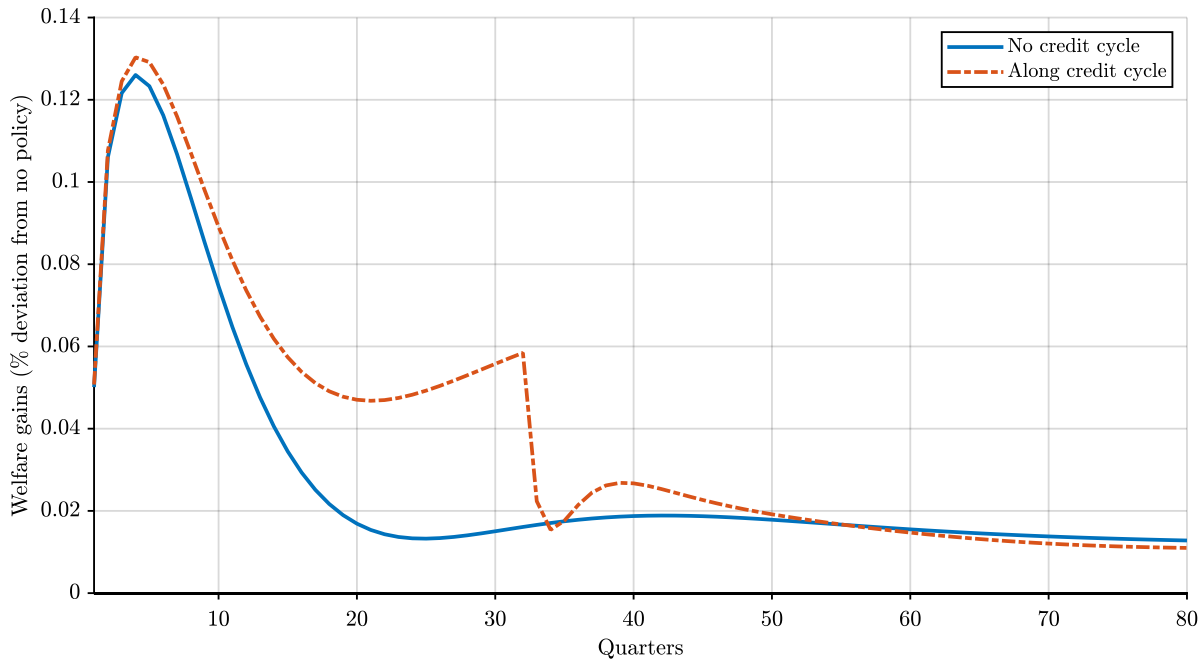
The tax is presented in annualized percentage points, so as its size can be compared to interest rates. Increasing the tax on CRE debt (and increasing the subsidy on final good producers' debt) decreases linearly the capital returns spread (upper right panel). This reduces the productivity-adjusted capital stock X_{ct} (bottom right panel). The shape of this function is concave, meaning that capital losses are minimal for small values of the tax: the productivity gains partially offset the distortion linked to higher taxes. Increasing CRE taxes also reduces labor (bottom left panel). However, the shape of the function is convex, meaning that the reduction in labor is larger for small values increase in the tax rate. This means that the consumption labor increases for small values of τ_h before decreasing afterwards, hence the inverted u-shape curves for welfare gains (upper left panel). Note that taking into account transition dynamics increases welfare gains. This is because when implementing the policy, households still benefits from the relatively high productivity-adjusted capital stock inherited from the initial steady state, while already benefiting from lower hours worked. In both case, the welfare-maximizing tax rate of CRE is around 80 basis points (77 without transition and 84 with transition). Given that at the steady state deposit interest rate is set at 2%, this represents a significant increase in CRE investors' borrowing costs, along with a sizable decrease in final good producers' borrowing costs (around 40 basis points). Note that the optimal policy does not fully close the capital returns spreads, which remains at 50 basis points in annualized rates.[12]

Although the resulting tax system is optimal from the perspective of $t = 0$, documenting the precise timing of welfare gains is important for political implementation. In addition, some macroeconomic context may be better suited to implement the transition. Although this would also be captured by the approach introduced above, such welfare gains are dwarfed by the computations over a long horizon. I thus compute period-by-period welfare gains of implementing the welfare-maximizing tax under two different scenarios: *(i)* there are no other shocks except the transition to new tax rates, *(ii)* the transition occur along the credit cycle introduced in Figure 6. More precisely, I assume that there is no redistributive tax implemented in period 0. In period 1, the new tax system is announced and implemented in both scenario. Therefore, agents realize that they are far away from the new steady state. They thus converge to this new steady state. In the second scenario, on top of this convergence, they adjust to the sequence of unexpected credit supply shocks. The welfare gains of the first scenario are hit by no other shocks, while that of the second scenario are computed against the no-policy economy hit by the same credit supply shocks.

[12]Note that welfare gains may appear small. This is partly because CRE investors and final good producers compete only for funds in my model. In reality, they can also compete for other resources, such as construction goods. I discuss the introduction of this channel in Section 6.

Figure 8 plots period-by-period welfare gains. As exposed previously, the transition is a source of welfare, with period-by-period welfare gains being greater than steady-state welfare gains in the two scenarios: households still benefits from the high productivity-adjusted capital stock while already benefiting from lower hours worked. However, welfare gains are larger if the policy is implemented during a credit boom. This is because capital returns spreads are even wider than at the steady state and misallocation is particularly salient. If the boom turns into a bust, welfare gains remain comparable to those of an implementation out of a credit cycle. Therefore, implementation in phases of credit expansion remains a good strategy. It makes clear that in the setup with exogenous credit supply shocks and credit cycles I study, the role of the policy is to maximize the efficiency of a credit boom, not to avoid a credit bust.

Figure 8: Period-by-period welfare gains of the transition to optimal constant tax rates



5.3 Time-varying tax rates

Once the economy has converged to optimal tax rates, policymakers may want to consider adjusting these taxes following the nature of the business cycle. Indeed, although misallocation is less salient in this new steady state, it remains time varying. In particular, it is increasing in times of credit expansion. If it is large enough, it may warrant policy action. Note that the fact the economy would particularly benefit from an increase in the tax on CRE debt τ_h during phases of credit expansion already hints in this direction. I thus compute period-by-period welfare gains of

implementing a time-varying tax rate in the context of positive credit supply shocks, such as the one studied in Figure 4. Specifically, I assume that the economy has reached the steady state with optimal tax rate set at $\tau_h = 0.84$ and that its time varying component follows a public rule:

$$\tau_{ht} = \tau_h + \phi_\tau \left(\hat{r}_{ht+1} - \hat{r}_{ct+1} \right) \tag{5.7}$$

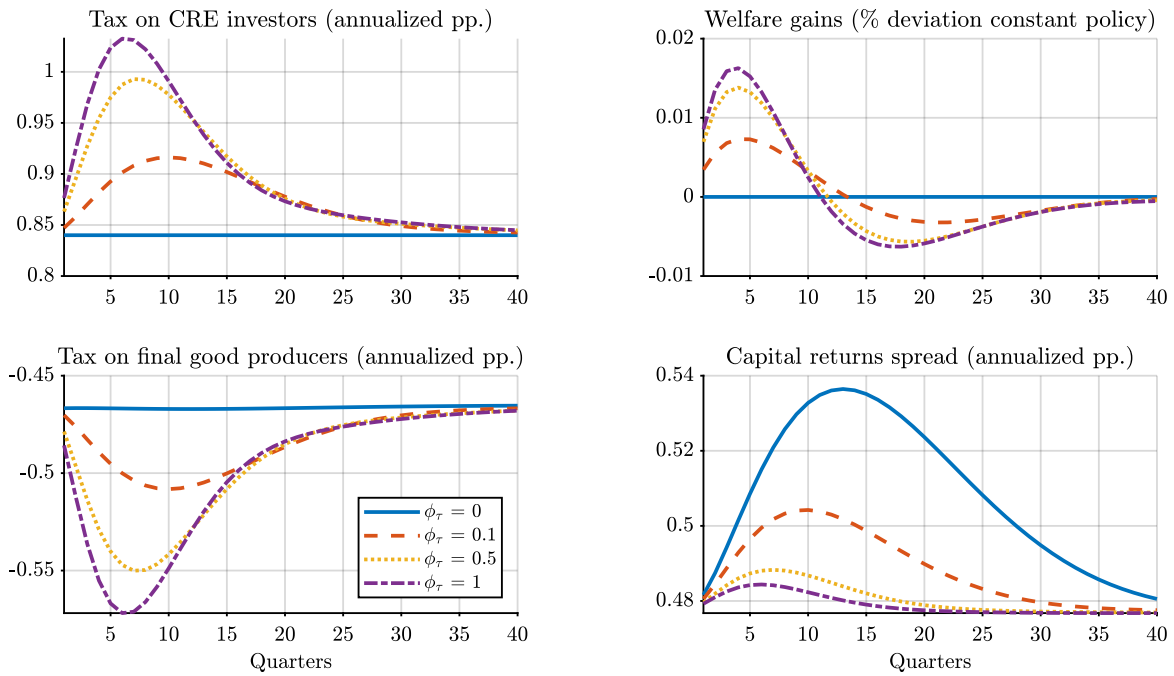
Where ϕ_τ controls for the responsiveness of the authority to deviations of the steady state returns' spread, expressed in log-deviation from steady state. As before, I let τ_{ct} adjust endogenously to maintain a balanced budget. Figure 9 presents these welfare gains for different values of ϕ_τ. In the absence of any policy, capital returns spread rise by 10 basis points at peak. Activating a time-varying rule enables to nearly mute fluctuations in capital returns' spread (bottom right panel), improving welfare (upper left panel). Note however that welfare gains are overall smaller than those of the transition to the optimal tax rates, approximately ten times smaller at peak. Adjusting these tax rates to economic fluctuations thus appear as a second-order problem compared to hitting optimal constant tax rates. Also note that the adjustment parameter does not need to bee large to be efficient: adopting $\phi_\tau = 0.5$ is sufficient to nearly close fluctuations in misallocation, welfare gains being marginal beyond this point. This result in a rather small adjustment on the tax rate τ_{ht}, which increases by 15 basis points at most (upper right panel), yielding less than 10 basis points in subsidies to final good producers (bottom right panel).

6 Robustness

The model I developed voluntarily abstracts from a number of empirically relevant features, so as to emphasize the main mechanisms behind my result. In this section, I consider alternative assumptions. Appendix B presents the full model, taking into account all the extensions discussed below.

Elasticity of substitution across capital types. This parameter is particularly complex to calibrate. I perform sensitive analysis of the exercise conducted in Figure 7 with respect to ω. This slightly changes the value of the optimal tax rate, but not the overall shape of the welfare function. The optimal tax is an increasing function of the elasticity of substitution, from 0.5% for low elasticity to 1% for high elasticity. The size of the welfare case is maximal for intermediate cases, such as the baseline calibration, where $\omega = 0$.

Figure 9: Impulse responses to a credit supply shock with time-varying policy



Imperfect rental market for CRE. In the baseline model, the CRE rental market is in perfect competition and the rental rate is equal to its marginal productivity. However, CRE goods are arguably imperfect substitutes, leading to market power and markups. Introducing this feature in my model dampens the misallocation problem: because CRE investors have market power, the rental rate is higher than the marginal productivity of capital, so CRE investors need to accumulate less capital to match households' discounted expected return. Nonetheless, final good producers also have market power, such that misallocation depends on *relative* markups. It is then not obvious whether markups are much higher on real estate rental market. For instance, Costa (2025) estimates that markups on the New York rental housing market range from 14% to 25%, roughly in line with the conventional 20% markup assumed in new-Keynesian models. Similarly, another source of rigidity make from rent adjustment costs, which in turn reduces volatility in capital misallocation. However, rents are (partially) indexed to inflation, such that the degree of rigidity is arguably low, and this mechanism is secondary.

Alternative borrowing constraint for final good producers. In section 2, I highlighted that non CRE investors do not frequently pledge their assets as collateral in their credit relationships. This suggests that the borrowing constraint these firms face may not be formulated in terms of assets, but rather in terms of income, in line with Drechsel (2023). I thus analyze an economy where CRE investors

face an asset-based constraint, while final good producers face an income-based constraint. [*Work in progress*]

Competition for construction goods. In the baseline model I analyze, capital stocks accumulated by both types of firms are purely sector-specific: there is thus no competition for any resource beyond funds. In practice, one may argue that both sectors compete for constructions goods. I thus introduce a construction sector that produces investment goods by combining labor and land, in fixed supply. CRE investors and final good producers thus compete for $I_t = I_{ct} + I_{ht}$ on a perfectly competitive with price q_t. Intuitively, this reinforces the misallocation problem, as the lower funding costs of CRE investors not only crowd out final good producers on financial markets, but on a physical markets. [*Work in progress*]

Constraint on dividend distribution. In the baseline model, there is no rigidity on dividend distribution, while in reality they are known to be rigid (Jermann and Quadrini, 2012). This is of importance for the dynamic behavior of the model, and in particular for credit cycles simulations, as dividends are allowed to be negative in crisis time. Implementing a lower bound on dividend distributions should intuitively lead to sharper drop of investment in crisis time, and in particular for CRE investors, as firms have to liquidate their assets so as to distribute dividends, thereby depreciating prices and further tightening the firms' borrowing constraint. [*Work in progress*]

7 Conclusion

This paper has shown how sectoral credit frictions create misallocation, focusing on the case of CRE investors. Empirically, these investors are particularly reliant on credit, as their assets are more easily pledgeable than those of other non-financial firms. Building on a 2-sector general equilibrium model, I then showed that this feature generates sectoral misallocation, especially in the face of positive credit supply shocks. This misallocation is not entirely inefficient, as it directs capital accumulation toward the least financially constrained sector, but nonetheless leaves room for policy intervention because the two types of capital are imperfect substitutes. Redistributive taxes on credit are welfare-improving, particularly when implemented during the expansion phase of a credit cycle.

The paper therefore argues that macroprudential authorities should consider applying sectoral credit policies that reallocate credit from CRE investors toward other productive non-financial sec-

tors. Increasing the interest rate charged to CRE investors by around 75 basis points would fund a 40-basis-point reduction in the interest rate charged to other non-financial firms. Such a policy would enhance welfare by improving the consumption-labor ratio of households, especially if implemented during a credit boom, when capital misallocation is particularly salient. This stands in contrast with current regulation, which favors low interest rates for CRE investors through lower capital requirements on CRE loans. While such policies are often justified by lower *microeconomic* losses in case of default for CRE investors, they fail to account for the inefficient *macroeconomic* capital allocation documented in this paper.

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A Figures and tables

Table 4: Liability structure - Independent firms

	Dependent variable:					
	Debt / total assets		Debt / debt and equity		Mortgage / bank credit	
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.210***	0.098***	0.393***	0.225***	0.169***	0.129***
	(0.0003)	(0.004)	(0.0004)	(0.007)	(0.0004)	(0.019)
CRE investor	0.328***	0.231***	0.264***	0.159***	0.413***	0.384***
	(0.014)	(0.008)	(0.015)	(0.006)	(0.022)	(0.024)
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Firm-level controls	No	Yes	No	Yes	No	Yes
Observations	4,295,127	3,288,056	4,295,127	3,288,056	931,374	832,823
R^2	0.039	0.231	0.021	0.292	0.015	0.023

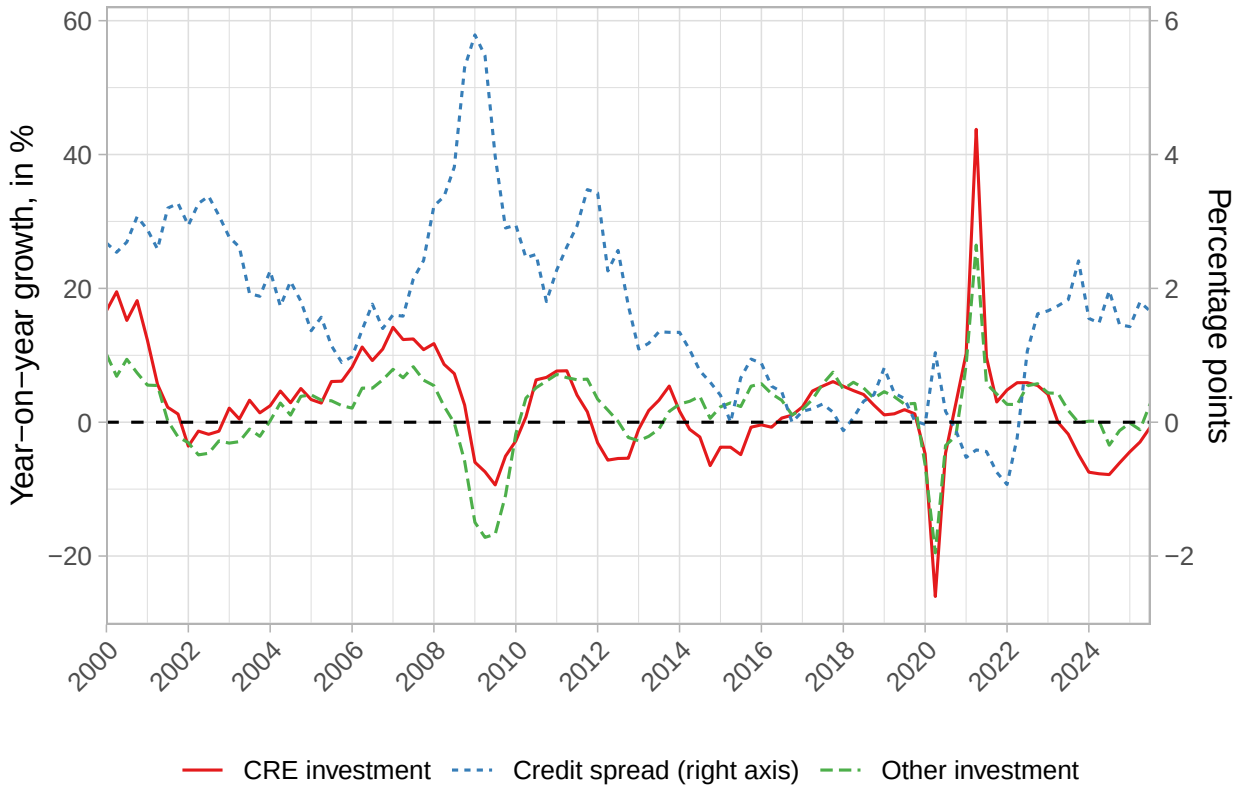
Note: *p<0.1; **p<0.05; ***p<0.01
Clustered standard errors, in parenthesis, at the year and firm levels.

Table 5: Credit growth and investment - Independent firms

	Dependent variable:			
	Fixed assets		Total assets in t+1	
	(1)	(2)	(3)	(4)
Credit	0.446***	0.427***	0.978***	0.946***
	(0.025)	(0.029)	(0.019)	(0.017)
Credit x CRE investor	0.229***	0.228***	0.222***	0.237***
	(0.025)	(0.027)	(0.030)	(0.034)
Sector x Year fixed effects	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Firm controls	No	Yes	No	Yes
Observations	4,055,022	3,187,987	3,640,582	2,824,454
R^2	0.405	0.439	0.389	0.479

Note: *p<0.1; **p<0.05; ***p<0.01
Clustered standard errors, in parenthesis, at the year and firm levels.

Figure 10: Credit spreads and sectoral investment



Source: INSEE, Refinitiv Workspace and Krippner (2013).

Table 6: Estimation results

	Prior			Posterior			
	Dist.	Mean	St. dev.	Mean	St. dev.	HPD inf	HPD sup
φ_c	norm	5.000	3.0000	2.340	0.9654	1.2834	3.3386
φ_h	norm	5.000	3.0000	1.315	0.2327	0.9564	1.6610
ρ_κ	beta	0.500	0.2000	0.891	0.0245	0.8514	0.9309
ρ_c	beta	0.500	0.2000	0.675	0.1392	0.4384	0.8926
ρ_h	beta	0.500	0.2000	0.745	0.0740	0.6230	0.8630
ρ_a	beta	0.500	0.2000	0.698	0.0621	0.5977	0.8008
σ_κ	invg	0.500	0.2000	0.581	0.0400	0.5141	0.6442
σ_c	invg	0.500	0.2000	4.351	2.5933	2.1844	6.5458
σ_h	invg	0.500	0.2000	2.963	0.6497	2.0670	3.8820
σ_a	invg	0.500	0.2000	3.582	0.2452	3.1864	3.9825

B Full model and equilibrium conditions

B.1 Final good producers

Production:

$$Y_t = X_{ct}^{\alpha}\Big(\exp(\zeta_{at})N_t\Big)^{1-\alpha} \tag{B.1}$$

$$X_{ct} = \left(\nu H_t^{\omega} + (1-\nu)K_{ct-1}^{\omega}\right)^{\frac{1}{\omega}} \tag{B.2}$$

Capital accumulation:

$$K_{ct} = (1-\delta)K_{ct-1} + \exp(\zeta_{ct})\left[1 - \frac{\varphi_c}{2}\left(\frac{I_{ct}}{I_{ct-1}} - 1\right)^2\right] I_{ct} \tag{B.3}$$

Borrowing constraint:

$$B_{ct} \leq \theta_c Q_{ct} K_{ct} \tag{B.4}$$

Budget constraint:

$$d_{ct} + (1+r_{t-1})B_{ct-1} + w_t N_t + r_{ht} H_t + I_{ct} \leq Y_t + B_{ct} \tag{B.5}$$

Objective:

$$\mathcal{V}_{ct} = \mathbb{E}_0 \sum_{j=0}^{\infty} \Lambda_{t,t+j} d_{ct+j} \tag{B.6}$$

Lagrangian

$$\begin{aligned}
\mathcal{L}_{ct} = \mathbb{E}_t \sum_{j=0} \Lambda_{t,t+j} \Bigg[& d_{ct+j} \\
& + \lambda_{ct+j}\Big(Y_{t+j} + B_{ct+j} - d_{ct+j} - (1+r_{t+j})B_{ct+j-1} - w_{t+j}N_{t+j} - r_{ht+j}H_{t+j} - I_{ct+j}\Big) \\
& + Q_{ct+j}\left((1-\delta)K_{ct+j-1} + \exp(\zeta_{ct+j})\left[1 - \frac{\varphi_c}{2}\left(\frac{I_{ct+j}}{I_{ct+j-1}} - 1\right)^2\right] I_{ct+j} - K_{ct+j}\right) \\
& + \mu_{ct+j}\Big(\theta_c Q_{ct+j} K_{ct+j} - B_{ct+j}\Big)\Bigg]
\end{aligned} \tag{B.7}$$

Equilibrium conditions:

$$1 = \lambda_{ct} \tag{B.8}$$

$$\lambda_{ct} = \mu_{ct} + \mathbb{E}_t m_{t,t+1} \lambda_{ct+1}(1+r_t) \tag{B.9}$$

$$Y_{nt} = w_t \tag{B.10}$$

$$Y_{ht} = r_{ht} \tag{B.11}$$

$$Q_{ct} = \mathbb{E}_t m_{t,t+1}\left[(1-\delta)Q_{ct+1} + \lambda_{ct+1} Y_{kt+1}\right] + \mu_{ct}\theta_c Q_{ct} \tag{B.12}$$

$$\exp(\zeta_{ct})Q_{ct} = \lambda_{ct} + \exp(\zeta_{ct})Q_{ct}\frac{\varphi_c}{2}\left(\frac{I_{ct}}{I_{ct-1}} - 1\right)\left(3\frac{I_{ct}}{I_{ct-1}} - 1\right) - \mathbb{E}_t m_{t,t+1}\exp(\zeta_{ct+1})Q_{ct+1}\varphi_c\left(\frac{I_{ct+1}}{I_{ct}} - 1\right)\left(\frac{I_{ct+1}}{I_{ct}}\right)^2 \tag{B.13}$$

$$K_{ct} = (1-\delta)K_{ct-1} + \exp(\zeta_{ct})\left[1 - \frac{\varphi_c}{2}\left(\frac{I_{ct}}{I_{ct-1}} - 1\right)^2\right]I_{ct} \tag{B.14}$$

$$\mu_{ct}\left(\theta_c Q_{ct}K_{ct} - B_{ct}\right) = 0 \tag{B.15}$$

$$d_{ct} + (1+r_{t-1})B_{ct-1} + w_t N_t + r_{ht}H_t + I_{ct} = Y_t + B_{ct} \tag{B.16}$$

$$X_{ct} = \left(\nu H_t^{\omega} + (1-\nu)K_{ct-1}^{\omega}\right)^{\frac{1}{\omega}} \tag{B.17}$$

$$Y_t = X_{ct}^{\alpha}\Big(\exp(\zeta_{at})N_t\Big)^{1-\alpha} \tag{B.18}$$

$$Y_{nt} = (1-\alpha)\exp(\zeta_{at})\Big(\frac{X_{ct}}{\exp(\zeta_{at})N_t}\Big)^{\alpha} \tag{B.19}$$

$$Y_{ht} = \nu(H_t/X_{ct})^{\omega-1}\alpha\Big(\frac{X_{ct}}{\exp(\zeta_{at})N_t}\Big)^{\alpha-1} \tag{B.20}$$

$$Y_{kt+1} = (1-\nu)(K_{ct}/X_{ct+1})^{\omega-1}\alpha\Big(\frac{X_{ct+1}}{\exp(\zeta_{at+1})N_{t+1}}\Big)^{\alpha-1} \tag{B.21}$$

B.2 CRE investors

Production:

$$H_{it} = K_{it-1} \tag{B.22}$$

Capital accumulation:

$$K_{it} = (1-\delta)K_{it-1} + \exp(\zeta_{ht})\left[1 - \frac{\varphi_h}{2}\left(\frac{I_{it}}{I_{it-1}} - 1\right)^2\right]I_{it} \tag{B.23}$$

Borrowing constraint:

$$B_{it} \leq \theta_h Q_{it}K_{it} \tag{B.24}$$

Demand constraint:

$$H_{it} = \left(\frac{r_{it}}{r_{ht}}\right)^{-\epsilon}H_t \tag{B.25}$$

Budget constraint:

$$d_{it} + (1+r_{t-1})B_{it-1} + I_{it} + \frac{v}{2}\left(\frac{r_{it}}{r_{it-1}} - 1\right)^2 H_t \leq r_{it}H_{it} + B_{it} \tag{B.26}$$

Objective:

$$\mathcal{V}_{it} = \mathbb{E}_t\sum_{j=0}^{\infty}\Lambda_{t,t+j}d_{it} \tag{B.27}$$

Lagrangian:

$$
\begin{aligned}
\mathcal{L}_{it} = \mathbb{E}_t \sum_{j=0} \Lambda_{t,t+j} \Bigg[& d_{it+j} \\
& + \lambda_{it+j} \left(r_{it+j} H_{it+j} + B_{it+j} - d_{it+j} - (1 + r_{t+j}) B_{it+j-1} - I_{it+j} - \frac{v}{2} \left(\frac{r_{it+j}}{r_{it+j-1}} - 1 \right)^2 H_{t+j} \right) \\
& + Q_{it+j} \left((1-\delta) K_{it+j-1} + \exp(\zeta_{ht+j}) \left[1 - \frac{\varphi_h}{2} \left(\frac{I_{it+j}}{I_{it+j-1}} - 1 \right)^2 \right] I_{it+j} - K_{it+j} \right) \\
& + \mu_{it+j} \left(\theta_h Q_{it+j} K_{it+j} - B_{it+j} \right) \\
& + \zeta_{it+j} \left(\left(\frac{r_{it+j}}{r_{ht+j}} \right)^{-\epsilon} H_{t+j} - H_{it+j} \right) \Bigg]
\end{aligned}
\tag{B.28}
$$

Equilibrium conditions:

$$1 = \lambda_{it} \tag{B.29}$$

$$\lambda_{it} = \mu_{it} + \mathbb{E}_t m_{t,t+1} \lambda_{it+1} (1 + r_t) \tag{B.30}$$

$$Q_{it} = \mathbb{E}_t m_{t,t+1} \left[(1-\delta) Q_{it+1} + (\lambda_{it+1} r_{it+1} - \zeta_{it+1}) H_{ikt+1} \right] + \mu_{it} \theta_h Q_{it} \tag{B.31}$$

$$\exp(\zeta_{ht}) Q_{it} = \lambda_{it} + \exp(\zeta_{ht}) Q_{it} \frac{\varphi_h}{2} \left(\frac{I_{it}}{I_{it-1}} - 1 \right) \left(3 \frac{I_{it}}{I_{it-1}} - 1 \right) - \mathbb{E}_t m_{t,t+1} \exp(\zeta_{ht+1}) Q_{it+1} \varphi_h \left(\frac{I_{it+1}}{I_{it}} - 1 \right) \left(\frac{I_{it+1}}{I_{it}} \right)^2 \tag{B.32}$$

$$\lambda_{it} H_{it} = \zeta_{it} \frac{\varepsilon}{r_{ht}} \left(\frac{r_{it}}{r_{ht}} \right)^{-\varepsilon-1} H_t + v \left[\lambda_{it} \frac{1}{r_{it-1}} \left(\frac{r_{it}}{r_{it-1}} - 1 \right) H_t - \mathbb{E}_t m_{t,t+1} \lambda_{it+1} \frac{r_{it+1}}{r_{it}^2} \left(\frac{r_{it+1}}{r_{it}} - 1 \right) H_{t+1} \right] \tag{B.33}$$

$$K_{it} = (1-\delta) K_{it-1} + \exp(\zeta_{ct}) \left[1 - \frac{\varphi_h}{2} \left(\frac{I_{it}}{I_{it-1}} - 1 \right)^2 \right] I_{it} \tag{B.34}$$

$$\mu_{it} \left(\theta_h Q_{it} K_{it} - B_{it} \right) = 0 \tag{B.35}$$

$$H_{it} = \left(\frac{r_{it}}{r_{ht}} \right)^{-\epsilon} H_t \tag{B.36}$$

$$d_{it} + (1 + r_{t-1}) B_{it-1} + I_{it} + \frac{v}{2} \left(\frac{r_{it}}{r_{it-1}} - 1 \right)^2 H_t = r_{it} H_{it} + B_{it} \tag{B.37}$$

$$H_{it} = K_{it-1} \tag{B.38}$$

$$H_{ikt+1} = 1 \tag{B.39}$$

B.3 Financial intermediaries

Balance sheet:

$$B_t = S_t + D_t \tag{B.40}$$

Leverage constraint:

$$S_t \geq \kappa_t B_t \tag{B.41}$$

Objective:

$$\Pi_{bt} = r_t B_t - r_{st} S_t - r_{dt} D_t \tag{B.42}$$

Equilibrium conditions:

$$B_t = S_t + D_t \tag{B.43}$$

$$S_t = \kappa_t B_t \tag{B.44}$$

$$r_t = (1 - \kappa_t) r_{dt} + \kappa_t r_{st} \tag{B.45}$$

$$\kappa_t = \kappa + \zeta_{\kappa t} \tag{B.46}$$

B.4 Households

Budget constraint:

$$C_t + D_t + S_t + \frac{\psi}{2}\left(\frac{S_t}{\bar{S}_t} - 1\right)^2 \bar{S}_t \leq w_t N_t + (1 + r_{dt-1}) D_{t-1} + (1 + r_{st-1}) S_{t-1} + d_{ht} + d_{ct} \tag{B.47}$$

Objective:

$$\mathcal{V}_t = \mathbb{E}_t \sum_{j=0} \beta^j \left(\frac{1}{1-\sigma} C_{t+j}^{1-\sigma} - \frac{\chi}{1+\eta} N_{t+j}^{1+\eta}\right) \tag{B.48}$$

Equilibrium conditions:

$$w_t = -\frac{\chi N_t^{\eta}}{C_t^{-\sigma}} \tag{B.49}$$

$$C_t^{-\sigma} = \beta \mathbb{E}_t C_{t+1}^{-\sigma} (1 + r_{dt}) \tag{B.50}$$

$$C_t^{-\sigma}\left[1 + \psi\left(\frac{S_t}{\bar{S}_t} - 1\right)\right] = \beta \mathbb{E}_t C_{t+1}^{-\sigma} (1 + r_{st}) \tag{B.51}$$

$$C_t + D_t + S_t + \frac{\psi}{2}\left(\frac{S_t}{\bar{S}_t} - 1\right)^2 \bar{S}_t = w_t N_t + (1 + r_{dt-1}) D_{t-1} + (1 + r_{st-1}) S_{t-1} + d_{ct} + d_{ht} \tag{B.52}$$

$$\bar{S}_t = \kappa B_t \tag{B.53}$$

$$m_{t,t+1} = \gamma \beta \mathbb{E}_t \frac{C_{t+1}^{-\sigma}}{C_t^{-\sigma}} \tag{B.54}$$

B.5 Exogenous processes

$$\zeta_{\kappa t} = \rho_\kappa \zeta_{\kappa t-1} + \varepsilon_{\kappa t} \tag{B.55}$$

$$\zeta_{ht} = \rho_h \zeta_{ht-1} + \varepsilon_{ht} \tag{B.56}$$

$$\zeta_{ct} = \rho_c \zeta_{ct-1} + \varepsilon_{ct} \tag{B.57}$$

$$\zeta_{at} = \rho_a \zeta_{at-1} + \varepsilon_{at} \tag{B.58}$$

B.6 Aggregation and market clearing

$$B_{ht} = \int_0^1 B_{it}\mathrm{d}i; \quad d_{ht} = \int_0^1 d_{it}\mathrm{d}i; \quad K_{ht} = \int_0^1 K_{it}\mathrm{d}i; \quad I_{ht} = \int_0^1 I_{it}\mathrm{d}i \tag{B.59}$$

$$r_{ht} = \left(\int_0^1 r_{it}^{\varepsilon-1}\mathrm{d}i\right)^{\frac{1}{\varepsilon-1}} \tag{B.60}$$

$$H_t = \left(\int_0^1 H_{it}^{\frac{\varepsilon-1}{\varepsilon}}\mathrm{d}i\right)^{\frac{\varepsilon}{\varepsilon-1}} \tag{B.61}$$

$$B_t = B_{ct} + B_{ht} \tag{B.62}$$

$$Y_t = C_t + I_{ct} + I_{ht} + \frac{v}{2}\left(\frac{r_{ht}}{r_{ht-1}} - 1\right)^2 H_t + \frac{\psi}{2}\left(\frac{S_t}{\bar{S}_t} - 1\right)^2 \bar{S}_t \tag{B.63}$$

C Proof of exogenous credit spreads

Combining the Euler equations of households for deposits and stocks:

$$1 + r_{st} = (1 + r_{dt})\left[1 + \psi\left(\frac{S_t}{\bar{S}_t} - 1\right)\right] \tag{C.1}$$

As the leverage requirement is binding:

$$1 + r_{st} = (1 + r_{dt})\left[1 + \psi\left(\frac{\kappa_t}{\kappa} - 1\right)\right] \tag{C.2}$$

Log-linearization around the deterministic steady state yields:

$$\hat{r}_{st} = \hat{r}_{dt} + \psi(1 + r)\hat{\kappa}_t \tag{C.3}$$

In addition, note that financial intermediaries' optimality condition can be expressed as follows:

$$\hat{r}_t = (1 - \kappa)\hat{r}_{dt} + \kappa\hat{r}_{st} \tag{C.4}$$

Combining these two equations and normalizing $\psi = \frac{1}{(1+r)\kappa}$:

$$\hat{r}_t - \hat{r}_{dt} = \hat{\kappa}_t \tag{C.5}$$